Management Science

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https://pubsonline.informs.org/journal/mnsc

MANAGEMENT SCIENCE
Vol. 71, No. 6, June 2025, pp. 4915–4941
ISSN 0025-1909 (print), ISSN 1526-5501 (online)

Do Consumers Respond to Social Movements? Evidence from Gender-Stereotypical Purchases After #MeToo

Clément S. Bellet,[a,*] **David Dubois,**[b] **Frédéric Godart**[b]

[a] Erasmus University Rotterdam, 3062 PA Rotterdam, Netherlands; [b] INSEAD, 77300 Fontainebleau, France
*Corresponding author
Contact: bellet@ese.eur.nl, https://orcid.org/0000-0002-8243-5969 (CSB); david.dubois@insead.edu (DD); frederic.godart@insead.edu, https://orcid.org/0000-0001-7530-8577 (FG)

Received: August 3, 2022
Revised: November 15, 2023; March 17, 2024
Accepted: April 21, 2024
Published Online in Articles in Advance: September 23, 2024

https://doi.org/10.1287/mnsc.2022.02352

Abstract. This paper explores whether and how consumers respond to global social movements challenging systemic discrimination and stereotypes. We examine the impact of the #MeToo movement on the market for products with stereotypical markers of femininity. Our analysis of high-frequency stockout and price data from a leading global fashion retailer spans from January 2017 to December 2018 and covers 32 countries, or 89% of the Organization for Economic Cooperation and Development (OECD) population. Using a triple-difference approach across time, countries, and products, we identify changes in product-level stockouts, corresponding to a drop in the purchase of stereotypically "feminine" items. Our findings suggest a sudden shift in demand, with no immediate adjustments in product assortments or pricing. We discuss potential driving mechanisms, as well as implications for product management and marketing in marketplaces where segmentation is gender based.

History: Accepted by Yan Chen, behavioral economics and decision analysis.
Supplemental Material: The online appendix and data files are available at https://doi.org/10.1287/mnsc.2022.02352.

Keywords: social movements • consumption • gender stereotypes • natural experiments • online retail • fashion

1. Introduction

Social movements—sustained collective action challenging existing norms and identities to drive societal change (Polletta and Jasper 2001, Kranton 2016)—have garnered increased visibility through social media. They have recently challenged stereotypes perceived as stigmatizing widely recognized identities, such as Black people for #BlackLivesMatter or women for #MeToo. Despite their well-established record in changing social beliefs and influencing public policies, less is known on whether and how these movements impact consumer markets.[1] We address this question by examining whether a global social movement like #MeToo can affect the purchase of products displaying stereotypical markers of femininity, and if so, why.

#MeToo or "Me Too" has become one of the most noticeable and defining social movements of the early 21st century.[2] Although the movement primarily focused on raising awareness of sexual harassment and assault, thereby resulting in higher reports of sex crimes (Levy and Mattsson 2023), it also challenged traditional gender roles and stereotypes (Hong and Zhang 2021). In particular, stereotypical views of masculinity (e.g., dominance, emotional suppression, protective paternalism) and femininity (e.g., passivity, vulnerability, objectification) can justify a culture of silence and sexism where sexual misconduct is either normalized or left unreported (Fiske and Glick 1995, Felmlee et al. 2020).

Stereotypes are generalized beliefs about attributes or characteristics traditionally associated with specific groups, including men and women (Coffman 2014; Bordalo et al. 2016, 2019; Coffman et al. 2019). Whereas the persistence of gender stereotypes in education and the labor market is well documented (Bertrand 2020, Coffman et al. 2021, Fischbacher et al. 2023), less is known on stereotyping within consumer markets. Firms frequently leverage gender to segment customers (Grohmann 2009) and resort to stereotyped customer representations to convey product information in a fast and effective way (Matta and Folkes 2005, Berger and Heath 2007). However, such strategies may backfire when they build on old clichés or perpetuate harmful beliefs about vulnerable groups (Lee et al. 2011, Bhattacharjee et al. 2014, Kim et al. 2023). For example, Victoria's Secret, a brand known for its reliance on stereotypical portrayals of femininity emphasizing sexual objectification and unrealistic body standards, faced notable backlash during the #MeToo movement (Thau 2018). The brand later decisively revamped its image by prominently featuring famous female soccer players, freestyle skiers, and tech investors (Maheshwari and Friedman 2021).[3]

The practice of targeting women by relying on stereotypical content is referred to as the "shrink it and pink it" strategy (Van Tilburg et al. 2015). Gender stereotypes in this context are often associated with traditional markers of femininity, such as colors (e.g., pink for girls versus black or blue for boys) and form (e.g., slimmer shapes, curvy lines emphasizing silhouette, versus bulkier, angular shapes symbolizing strength). The current paper primarily focuses on a product category closely tied to women's identity expression: the footwear market. Although we also analyze other product categories for robustness, women's shoes, when stripped of their stereotypical markers of femininity, become almost indistinguishable from men's shoes, making this category particularly suitable for the analysis (Belk 2003, Dilley et al. 2015).[4]

Using data from a leading global fashion retailer, we construct a unique and high-frequency panel that tracks the availability of various shoe sizes at the product level. These data provide a daily record of out-of-stock sizes for a total of 1,718 women's shoe models and 1,275 men's shoe models, all of which were accessible to customers in 32 Organization for Economic Cooperation and Development (OECD) countries (i.e., 89% of the total population of OECD countries). The data span from January 2017 to December 2018.

We categorize products as gender stereotypical based on the presence of traditional markers of femininity and validate this definition through an international survey of customers. To gauge the level of exposure to the #MeToo movement on a country-by-country basis, we employ a combination of Twitter data, Google search data, and national news headlines. The primary method of analysis relies on a triple-difference estimator, which involves comparing the number of out-of-stock sizes for products with stereotypical (versus nonstereotypical) markers of femininity in countries exposed to the #MeToo movement (versus unexposed countries) during the weeks following the movement's start (versus the period preceding it).

The findings reveal compelling evidence of a decline in the market for gender-stereotypical products in the wake of the #MeToo movement. In comparison with nonstereotypical products, we observe a substantial drop of about 25% in the number of out-of-stock sizes for products featuring stereotypical markers of femininity during the six weeks following October 15, 2017. Our results indicate a shift away from the most stereotypically feminine items, rather than a move toward more neutral or masculine alternatives. In our context, this effect is akin to imposing an 8% "sales tax" on stereotypical products. Importantly, the results remain robust when accounting for country-specific seasonality controls, different measures of country-level exposure to #MeToo, alternative approaches to operationalize stereotypical markers, and variations in the time window used to estimate the effect.

Other factors tied to consumer behavior preceding the #MeToo movement are unlikely to drive the effect. Exposed and unexposed countries exhibited similar trends in searches related to sexual harassment and in the relative frequency of stockouts for stereotypical versus nonstereotypical products before the emergence of #MeToo. Additionally, we conduct placebo tests using alternative starting points for the #MeToo movement, none of which yield significant estimates. The lack of effects when using men's products as an alternative category, or when analyzing out-of-stock sizes from a year later, confirms that it is unlikely the effect stems from more general trends in footwear.

An important aspect of the empirical setting is the ability to document changes in adherence to consumer stereotypes within an expenditure category that encompasses feminine and more neutral or masculine gender associations. However, we explore the potential generalizability of these results within exclusively feminine product categories. Although we lack survey-based measures of stereotypical femininity in these cases, we employ the color pink or red as a universally recognizable indicator of stereotypical femininity. The study extends to three additional women's product categories (lingerie, dresses, and handbags), revealing consistent reductions in stockouts for pink or red items, albeit with varying effect sizes.

In the final section of the paper, we delve into various channels and elaborate on potential behavioral mechanisms underlying the observed effects. Firstly, although the estimated impact of #MeToo on product stockouts could theoretically align with both demand-side effects (such as a shift in consumer preferences) and supply-side effects (like retailers altering their product assortments or pricing strategies), the findings strongly support a demand-side channel. We scrutinize whether the retailer responded swiftly to #MeToo by either discontinuing stereotypical products or introducing less stereotypical items into their offerings. However, we find no evidence of sudden changes in product assortments that could account for the observed effect. Notably, there was a reduction in the number of stereotypical products offered by the retailer after #MeToo, but this reduction pervaded both exposed and unexposed countries. In contrast to inventory allocation decisions which take time to process, prices can adjust more rapidly. We also investigate whether the retailer increased the prices of stereotypical products shortly after #MeToo. We find no evidence of the retailer altering its pricing for these products in response to the movement.

Secondly, we explore whether the documented effect aligns with the timing of daily tweets related to #MeToo on social media and the subsequent surge in online searches about sexual harassment cases. The analysis reveals that the number of daily tweets about #MeToo in a given day within a given country correlates with fewer stockouts for gender-stereotypical products in that same day. We find a similar effect for higher day-to-day searches on sexual harassment within countries, but this effect becomes apparent only after the advent of #MeToo.[5] This suggests that the effect cannot be solely explained by increased awareness of sexual assault and misconduct. Instead, it appears to stem from an activated behavioral link between greater salience of harmful behaviors and the perpetuation of gender stereotypes in society.

We consider three potential behavioral mechanisms accounting for the drop in customer adherence to stereotypes. First, #MeToo may have served as a reminder to consumers of identity-related threats, which encompass concerns about safety and the stigma associated with wearing stereotypical products (e.g., victim blaming). In other words, women may have felt a heightened sense of vulnerability and concern about wearing stereotypical items in public immediately after #MeToo, especially if these products are perceived as inviting men's attention or supportive of harmful stereotypes.[6] A second explanation relates to women's empowerment. If stereotypical markers of femininity are seen as catering to men's preferences rather than meeting women's needs, women who felt empowered by #MeToo might have chosen to purchase fewer stereotypical products. Finally, a third explanation involves values alignment. By sharing harassment cases, #MeToo challenged fundamental aspects of sexism and patriarchal societies, calling for a shift in established gender roles. Dissociating from stereotypical markers may thus signal a deeper desire to support the movement's values in its endeavor to combat sexism and deconstruct stereotypical views about gender. We discuss each explanation through additional empirical tests and the results of an online experiment. The experiment confirms the negative impact of #MeToo on preferences for gender-stereotypical products.

This research offers two key contributions. First, although the management literature has recently started to incorporate insights from social identity theory (Coffman et al. 2021, Del Carpio and Guadalupe 2021, Fischbacher et al. 2023), there is a lack of empirical evidence on whether and how social movements play a role in consumer markets.[7] This gap in the literature is primarily due to the inherent complexity of studying such effects in consumer settings. First, these movements can exert an impact on both sides of the market, affecting not only consumer demand but also firms' production decisions and thus the range of products available to customers (Godart et al. 2023). Second, consumption data, such as surveys, often lack the necessary granularity in terms of time, space, and product description. Whereas other recent studies have looked at the causal effect of #MeToo on outcomes directly related to harassment such as reporting of sexual crimes (Levy and Mattsson 2023), quitting rates from toxic work environments (Batut et al. 2021), or women hires in Hollywood (Luo and Zhang 2021b), our work uniquely identifies a novel effect indicative of a shift in consumer attitudes toward gender stereotypes.

Second, we contribute to an important body of research in economics and marketing that explores the social determinants of consumer preferences, including factors such as culture or social identity (Benjamin et al. 2010, Atkin 2016, Yoganarasimhan 2017, Atkin et al. 2021). A distinctive feature of this study is the integration of the growing literature on gender stereotypes into our understanding of consumer markets. Adding to a rich set of online and field studies on gender stereotyping (Coffman 2014; Bordalo et al. 2016, 2019; Coffman et al. 2019, 2021), we formulate a novel argument on how social movement can alter adherence to gender stereotypes. Whereas the bulk of past efforts focused on gender stereotypes in labor or education markets, our work complements a smaller body of laboratory-based studies in consumer research (Lee et al. 2011, Kim et al. 2023) by providing field-level evidence of gender stereotyping avoidance in consumer markets.

The findings carry significant implications for managerial practices inspired by traditional segmentation strategies. Firms that heavily rely on stereotypical markers to target women customers may experience more significant challenges in the wake of identity-based social movements, in contrast to companies offering more neutral assortments. In the long run, social movements like #MeToo can induce shifts in the supply side, influencing stereotypical beliefs as producers invest in less stereotypical consumption norms. As Victoria's Secret newly appointed CEO Martin Waters told the *New York Times* in 2021, "we needed to stop being about what men want and to be about what women want" (Maheshwari and Friedman 2021). Further research could delve into the interplay between team diversity within firms and the types of goods (or advertising) they produce. The contraction of the market for gender-stereotypical products, as highlighted in the study, may for instance explain why more gender-equal companies witnessed an increase in their firm value following #MeToo (Lins et al. 2020). Although these insights go beyond the scope of this paper, they underscore the need to reevaluate traditional gender-based product segmentation and targeting strategies. Such strategies may have diminished in effectiveness and societal acceptability in the post-#MeToo era.

The remainder of this paper is organized as follows. Section 2 provides a theoretical background and overview of the literature. Section 3 describes the data. Next, Section 4 discusses the empirical setting and estimation procedure. The main results are presented in Section 5. Section 6 discusses channels and potential behavioral mechanisms. Finally, Section 7 concludes and proposes directions for future research.

2. Theory and Background

2.1. Identity-Based Social Movements

Social movements are a form of collective action devised to influence social outcomes such as regulations, beliefs, or behaviors (Tilly and Wood 2020). Recent years have witnessed the rise of digitally native movements (e.g., #MeToo, #BlackLivesMatter) that challenge systemic discrimination and harmful stereotypes against specific identities. These movements combine traditional modes of action (such as mass gatherings) with social media approaches (e.g., the use of hashtags) and online communication as a way to raise awareness on the victimization of marginalized groups. Their focus on identities (e.g., women, ethnic minorities) is encouraged, in part, by new digital means of interactions that help to challenge existing power structures and give rise to identity claims (Castells 2011).

Such social movements can be profound enough to instigate enduring social transformations. For example, Agarwal and Sen (2022) analyze the influence of #BlackLivesMatter protests on teachers' requests for antiracist educational materials in U.S. public schools. They find a notable uptick in requests for such resources, including books that highlight marginalized communities. A study by Garg et al. (2018), utilizing word embeddings trained on a century's worth of textual data, highlights important shifts in gender biases linked to specific occupations and adjectives (e.g., transitioning from "hysterical" to "emotional") following the women's movement of the 1960s. Although gender norms and stereotypes are remarkably persistent and resistant to change (Bertrand 2020, Eagly et al. 2020), they can be mitigated and weakened when individuals are exposed to powerful gender-inclusive interventions (Dahl et al. 2021).

After becoming the most widely searched term globally in October 2017, #MeToo progressively gave rise to many spinoffs around the world, yielding important attitudinal and behavioral changes. It notably led to a major rise in reporting sexual crimes to the police (Levy and Mattsson 2023), women being less accepting of toxic masculinity at work (Batut et al. 2021), greater rewards for more gender-equal companies in the form of greater excess returns (Lins et al. 2020), and women writers being hired more frequently in Hollywood (Luo and Zhang 2021b). For instance, Luo and Zhang (2021a) document fewer stereotypical feminine characters in movies after #MeToo, all of which support #MeToo's role as a transformative social movement challenging gender norms and stereotypes.[8] There is little empirical evidence, however, on whether and how customers respond to global social movements that challenge systemic discrimination and stereotypes against specific social identities.

2.2. Gender Identity Expression and Consumption

Consumers invest in products and brands as a means to express various facets of the self, or the groups they wish to be part of (Bhattacharjee et al. 2014, Atkin et al. 2021). These facets encompass a wide spectrum, including but not limited to gender, age, ethnicity, social roles (e.g., CEO versus factory worker), experience-based affiliations (e.g., alumni), and value-based community memberships (e.g., political ideology and religion). Identity expression motives become particularly pronounced when consumption is observable by others (Amaldoss and Jain 2005, Bursztyn et al. 2018, Bellet 2024), with clothing and footwear, being highly visible forms of consumption, drawing special attention (Heffetz 2011).

Despite contemporary efforts to blur gendered distinctions in product design (Reis et al. 2018), gender identity remains a pivotal organizational principle within the fashion industry and the broader consumer goods landscape (Aspers and Godart 2013). The strong link between gender identity and consumer markets is exemplified by Bertrand and Kamenica (2023), who accurately infer a person's ascribed gender based on their consumption patterns. This inference is facilitated by the segmentation of customer audience by gender, which leads to the presence of highly gender-specific brands (e.g., Victoria's Secret), products (e.g., makeup), or media consumption (e.g., *People* magazine).

Gender-based market segmentation involves the categorization of a market or target audience into distinct groups based on one or more of their most representative traits. This prevalent practice serves businesses by tailoring their products to align with consumers' aspirations for gender identity expression. However, it often relies on stereotypical concepts of masculinity and femininity, as demonstrated by strategies such as the "shrink it and pink it" approach (Van Tilburg et al. 2015).

Indeed, elements of design, including proportions, shapes, lines, and color choices, have traditionally been associated with either masculinity or femininity (Simonson and Schmitt 1997, Alexander 2003, Dilley et al. 2015). The use of the color pink or red to denote products for girls is a widely recognized instance of such association (Moss et al. 2006, Karniol 2011, Labrecque and Milne 2012). In a recent study investigating the "pink tax," Moshary et al. (2023) found that color was the most frequently cited attribute for categorizing the gender target from product images.[9] This is consistent with findings from large image recognition systems (Schwemmer et al. 2020). Additionally, curvy lines and slim, round shapes often associated with fragility are traditionally marketed to women, whereas bulkier, angular shapes symbolizing strength are typically associated with products targeted at men (Moss et al. 2006, Lieven et al. 2015). Within global contemporary footwear, heels—footwear that raises the heel significantly higher than the toes, emphasizing silhouette or gait—have also emerged as a distinctive "feminine" style at least since the 19th century (Morris et al. 2013, Van Tilburg et al. 2015). Heels represent a durable, recognizable pattern of aesthetic choices

(Godart 2018), with high popularity and cultural impact (Semmelhack 2008).[10]

Although gender-based market segmentation and targeting can be a tool to facilitate customers' self-expression, it is not without its challenges. These practices may not adapt fast enough to changes in customer beliefs about gender, and inadvertently diminish consumers' perception of agency in identity expression (Bhattacharjee et al. 2014, Kim et al. 2023). Relying on conventional markers of femininity can also be viewed as perpetuating old clichés and more harmful gender stereotypes linked to norms of physical attractiveness and body types (Tartaglia and Rollero 2015) or the sexual objectification of women (Labrecque and Milne 2012, Morris et al. 2013, Wright 2017).

2.3. Dissociation from Gender Stereotypes

We identify three possible mechanisms through which identity-based social movements may influence the purchase of stereotypical products: (i) the perception of identity-related threats, (ii) the empowerment of one's identity, and (iii) alignment with core values.

First, identity-based social movements often emerge as a reaction to traumatic events (e.g., the killing of George Floyd, Harvey Weinstein's sexual assaults) that highlight systemic discriminations experienced by identity groups. By shedding light on these events, social movements may increase the salience of identity threats. A first, immediate form of identity threat includes one's sense of safety (e.g., victimization, the fear of harassment or abuse). However, identity threats also pertain to situations where individuals believe their actions (e.g., how to dress) might confirm negative stereotypes associated with their identity group (i.e., "stereotype threat" (Spencer et al. 2016, Kim et al. 2023)), encouraging negative judgments from others (e.g., "women are fragile, weak, and powerless") or the social stigma resulting from those judgments (e.g., victim blaming). When faced with an identity threat, individuals tend to turn away from stereotypical behaviors attached to the threatened identity in order to minimize or avoid the threat (Ethier and Deaux 1994, White and Argo 2009, Lee et al. 2011, Kim et al. 2023). For instance, Lee et al. (2011) show that when women customers believe that an outgroup transaction partner (i.e., men) will view them through the lens of a stereotype (e.g., a gender-STEM stereotype), they are less likely to purchase products aligned with the stereotype (e.g., financial services where math skills are seen as central).[11] Similarly, Kim et al. (2023) show that identity appeals can backfire when they are associated with a stereotype-evoking appeal because customers do not want to feel pushed into a particular group ("categorization threat").[12]

A second potential mechanism centers on identity empowerment. As a way to cope with an identity threat or traumatic event, social movements can lead to empowerment by fostering group self-esteem (or status) and collective identity (Stryker et al. 2000). If social norms and stereotypes primarily reflect the expectations or preferences of the dominant group (e.g., men) rather than addressing the needs of the minority group (e.g., women), empowerment may lead members of the previously dominated group to depart from those norms and meet their own identity-related needs instead. For example, women might opt for clothing and accessories that prioritize comfort and practicality, and depart from traditional and restrictive gendered fashion choices.[13]

A third possible mechanism involves a more direct association with the values championed by the social movement itself. Identity-based social movements advocate broader political values, which can in turn lead to a form of "political consumerism" (i.e., consumers "voting" with their wallets (Stolle et al. 2005)). In other words, political views or beliefs can shape customer preferences and purchase behavior (Schoenmueller et al. 2023). Recent empirical work on the topic examines whether consumers intentionally cease purchasing certain products or brands (boycotts) or begin buying them (buycotts) for political reasons (Liaukonytė et al. 2023). However, unlike boycotts or buycotts, which explicitly target a specific brand or product in their communication, social movements tend to center their efforts on a set of core values. As for #MeToo, these values include the fight against sexism and the deconstruction of traditional gender roles (e.g., what should be defined as feminine), which in turn can lower customer adherence to stereotypes. They can diffuse throughout the population, shifting societal attitudes closer to the principles upheld by the movement.

In summary, a dynamic interplay exists between social movements, identity expression, and consumer behavior. In the rest of the paper, we describe the data and empirical approach, which primarily aims to assess whether customers responded to #MeToo by altering their consumption of gender-stereotypical products, and offer a magnitude of the effect.

3. Data

We conducted a comprehensive data collection across multiple countries. Initially, we leveraged high-frequency data on stockouts and prices of an exhaustive collection of 1,718 women's shoes (along with 1,275 men's shoes) from a leading global fashion retailer in 32 OECD countries between January 1, 2017, and December 31, 2018. To guide the categorization of products as "stereotypical" or "nonstereotypical," we gathered 36,000 product ratings from a diverse set of survey participants representing six OECD countries (Italy, France, Portugal, United States, United Kingdom, and Mexico). Additionally, to gauge #MeToo's influence at the country level, we collected data on (i) #MeToo tweets per million users, (ii) the number of news articles in national newspapers related to #MeToo, and (iii) Google search volumes on topics concerning sexual harassment.

3.1. Retailer Data

The main data set comprises millions of web-scraped data points capturing daily information on product listings from one of the largest global fashion retailers, with annual revenues above 20 billion euros.[14] The retailer sells trendy and affordable apparel and footwear under its own brand label, both online and in local stores. The brand caters to a diverse audience worldwide, appealing to both men and women mostly between the ages of 18 and 40 with a midrange income.

The data allow us to recover daily information on out-of-stock product sizes and product prices between January 2017 and December 2018 for 32 OECD countries.[15] Over this two-year period, the retailer offered a selection of 1,718 shoe models within its women's section on the platform.[16] For each shoe model, information such as color (black, blue, pink, red, etc.), price (in U.S. dollars), shoe style (e.g., heels and pumps, platforms and wedges, sneakers, sandals, boots, or lace-up shoes), and heel height measured in inches is recorded. About 30% of women's shoes belong to the "heels and pumps" category, 66% have heels (heel height higher than 1 inch), 22% have high heels (heel height higher than 3 inches) and about 10% are either pink or red (see Table S1 in the Online Appendix).

The original data track online product purchases by capturing any daily changes in out-of-stock size information for each possible size of each product in each country. They also track product price changes over the same period. Hence, the data reflect national demand and supply through variations in product size availability and product pricing on the retailer's website. Although daily stockout size information is binary (0 = product size unavailable; 1 = product size available), each product offers eight possible sizes, so the product-level stockout measure ranges between zero and eight. Nearly 400 shoe models can be purchased each month and remain available on the website for about four months. From the full [product × size × day] matrices, we construct a high-frequency measure of product-level stockout within each country, which reflects the daily number of out-of-stock shoe sizes per product. This measure ranges between zero and eight (mean = 2.11; SD = 2.71) and varies considerably within product-country over time (see Figure S1 in the Online Appendix; SD = 1.96).

Past studies, which have compared sales data with occurrences of product stockouts to estimate demand, consistently demonstrate a strong positive correlation between predicted sales and the frequency of product stockouts (Conlon and Mortimer 2013). In retail, when a product is not selling well, there is less pressure on the available stock, leading to reduced instances of stockouts across various sizes. In contrast, in-demand products tend to experience stockouts more frequently as they quickly sell out their available inventory. Two robustness checks confirm that the stockout measure effectively captures (unexpected) country-level sales shocks (see Online Appendix A). First, product-level stockouts exhibit regular variations consistent with industry seasonality (see Figure S2 in the Online Appendix). Notably, these peak at the start of winter sales (January) and summer sales (July) as well as during Black Fridays, where we observe a clear negative correlation with average product prices. Second, the data reliably capture product-level stockout variations specific to each country. In particular, daily country-level online searches for "Black Friday" (November 24, 2017) and corresponding drops in average product prices coincide with a spike in the stockout measure in countries that have a Black Friday. However, no such variations in online searches, product price, or stockout are observed in countries that do not have a Black Friday (see Figure S3 in the Online Appendix).

3.2. Gender-Stereotypical Products: Measurement

We operationalize gender stereotypes in product design by focusing on attributes traditionally associated with femininity, particularly related to forms and colors. From the literature review in Section 2, we pinpointed three specific markers of femininity within the data set: (i) heels and pumps, representing slimmer and exclusively feminine shoe shapes; (ii) heel height, measured in inches, also linked to forms; and (iii) the color pink or red.

When combined, these three markers create noticeable visual distinctions that influence how a women's shoe product is perceived in terms of its degree of "femininity." To exemplify this, Figure 1 showcases six products available in the women's wear section of the retailer, based on the presence or absence of stereotypical markers of femininity. This selection includes a product featuring all three markers (a pink high-heel pump), a product with two markers (a red high-heel platform), and two products with just one of the three "feminine" markers (a yellow flat pump and a pink flat boot). Additionally, we present two products without either marker (a green sneaker and a black lace-up).

To validate the operationalization of "stereotypical" markers, we conducted a survey involving an international sample of 1,200 respondents from six OECD countries (Italy, France, Portugal, United States, United Kingdom, and Mexico). Respondents were presented with product images similar to those featured in Figure 1 and were asked to assess the style of these products on a gendered scale. They answered the question "How would you describe this style of shoes? Very masculine

Figure 1. (Color online) Examples of Women's Products Sold on the Retailer's Website

Notes. (a) Pink high-heel (SI = 7). (b) Red platform (SI = 6). (c) Pink flat boot (SI = 5). (d) Yellow pump (SI = 5). (e) Green sneaker (SI = 3). (f) Black lace-up (SI = 3).

(1), Masculine (2), Somewhat masculine (3), Neutral (4), Somewhat feminine (5), Feminine (6), Very feminine (7)."[17]

The survey confirms that among the shoe color attributes, pink and—to a lesser extent—red are the two most stereotypically "feminine" colors (whereas black and brown are the least feminine (see Figure S5 in the Online Appendix). In terms of type of shoes, pumps are the most "feminine" style and lace-up shoes or sneakers the least feminine. Finally, heels (and in particular high heels) are perceived as significantly more feminine than flat shoes. We look for heterogeneity by respondent's gender (see Figure S6 in the Online Appendix) and country of residence (see Figure S7 in the Online Appendix) but find no evidence of significant differences along those dimensions. Although respondents may differ in their attitudes toward gender stereotyping, those results suggest that they share very similar views in terms of which product markers tend to be traditionally associated with women (versus men).

Using the estimates from the survey, we can calculate a predicted stereotypical index (SI) for each product in the data set. To illustrate, we display the estimated (rounded) SI value on the one-to-seven scale for each product image in Figure 1. The scale ranges from "very feminine" (SI = 7) for pink high heels to "somewhat feminine" (SI = 5) for yellow pumps and "somewhat masculine" (SI = 3) for green sneakers or black lace-ups. Although our primary analysis relies on the presence (or absence) of stereotypical markers, we replicate our results using the predicted stereotypical index as an alternative way to measure stereotypical products in the robustness section of the paper. For an in-depth explanation of the SI's construction, please refer to Online Appendix C.

3.3. Country-Level Exposure to #MeToo

#MeToo is a worldwide social movement with varying degrees of influence across countries. Having started on October 15, 2017, on Twitter,[18] it triggered substantial online activity extending beyond social networks. Moreover, it reached an even broader audience through extensive coverage in major national news outlets.

We create a composite index to gauge the level of country-level exposure to the #MeToo movement. This index is formed by gathering data from three distinct and complementary sources. Firstly, we collect all tweets containing MeToo-related hashtags (#MeToo or its local equivalent in the target country's language) starting from Alyssa Milano's initial tweet on October 15, 2017, and extending until November 2018. This comprehensive data set encompasses approximately 12.6 million tweets over a span of more than a year, with around 20% of these tweets concentrated within the initial six weeks following October 15, 2017. These tweets are written in various languages and originate from diverse locations, with English-written tweets being the most prominent (59% of the data set). To ascertain the tweet locations, we rely on user-supplied information, allowing us to geotag approximately 30% of the tweets.[19] Restricting the data set to tweets linked to OECD countries leaves us with a total of 2.5 million tweets. We follow Levy and Mattsson (2023) and standardize the measure of localized tweets by the number of Twitter users within each country.[20]

The Twitter-based measure, although valuable, may not fully capture the impact of #MeToo on segments of the population who are not active on social networks. To address this, we enhance this initial measure with an additional metric: the total count of news headlines related to #MeToo published in major national news outlets within the six weeks following October 15, 2017, for each of the 32 countries in the study. We use Factiva, a platform that offers access to around 25,000 international sources from 159 countries, available in 22 languages. We retrieved the number of articles published in print media that mentioned the term "#MeToo." To ensure that we captured local news accurately, the data set includes news headlines mentioning #MeToo or its local equivalent in the target country's language. In countries with more demand for news, the production of news articles may also be larger. To account for variations arising from differences in the size of readership markets across countries, we normalize this measure by dividing the number of #MeToo headlines by the country's population, expressing it as the number of #MeToo headlines per million residents.[21]

One inherent limitation of relying on national news outlets is their potential inability to promptly reflect online activities. Editorial considerations might also lead them to prioritize other national subjects over the #MeToo movement. This can result in an underestimation of the actual exposure to the movement. Twitter metrics, although addressing some of these issues, come with their own limitations. For instance, tweets are public, and some individuals interested in the movement may choose not to publicly display their support, potentially leading to an underestimation of the movement's true impact.

To address these limitations and offer a more direct measure of #MeToo exposure, we introduce an alternative approach: tracking search queries related to sexual harassment. We collect daily, country-level normalized web search volumes on Google Trends[22] for the topic "sexual harassment" from January 1, 2017, to December 31, 2018. By collecting searches at the topic level, we can capture a broader set of search terms associated with sexual harassment in a specific country, considering searches in local languages or related to names and cases associated with sexual harassment.[23] As a benchmark, we also gather normalized web search volumes for the topic "gender equality," which is linked to women's rights more broadly. The search-based measure of country-level exposure evaluates the increase in search interest in the topic of sexual harassment during the six weeks following the start of the #MeToo movement, compared with the six weeks preceding it.

Considering the advantages and disadvantages of each metric, we employ a composite index as the primary measure of exposure to #MeToo, although we also show results using each metric individually for robustness. We normalize each of the three #MeToo exposure metrics on a scale from zero to one. We calculate the average of these three metrics to derive the #MeToo exposure index. Figure 2 plots the resulting distribution of the index across OECD countries. Nations with low or minimal exposure to #MeToo, such as Lithuania, Japan, Turkey, or Mexico, are considered unexposed. In contrast, countries with substantial exposure encompass Sweden, France, the United States, and Finland.[24]

4. Empirical Setting

4.1. A Triple-Difference Estimator

Assuming that exposure to #MeToo is perfectly randomized across buyers, we could estimate an average treatment effect (ATE) of #MeToo. However, #MeToo exposure is not random. For instance, some consumers may be more or less exposed to #MeToo depending on their access to or consumption of media online and offline. Rather than the ATE, our goal is therefore to estimate an average treatment effect on the treated (ATT). We can define the ATT as

$$E(S_{i1} - S_{i0} | T = 1) = \beta,$$

where S_{i1} is the number of sales of product i in the treatment condition, S_{i0} is the number of sales of product i from the control condition, and $T = 1$ indicates that consumers of product i were among those who were exposed to #MeToo. The identification challenge arises from our lack of knowledge of counterfactual sales S_{i0}, namely, how many units of product i would have been purchased by the treated consumers in the absence of a #MeToo shock.

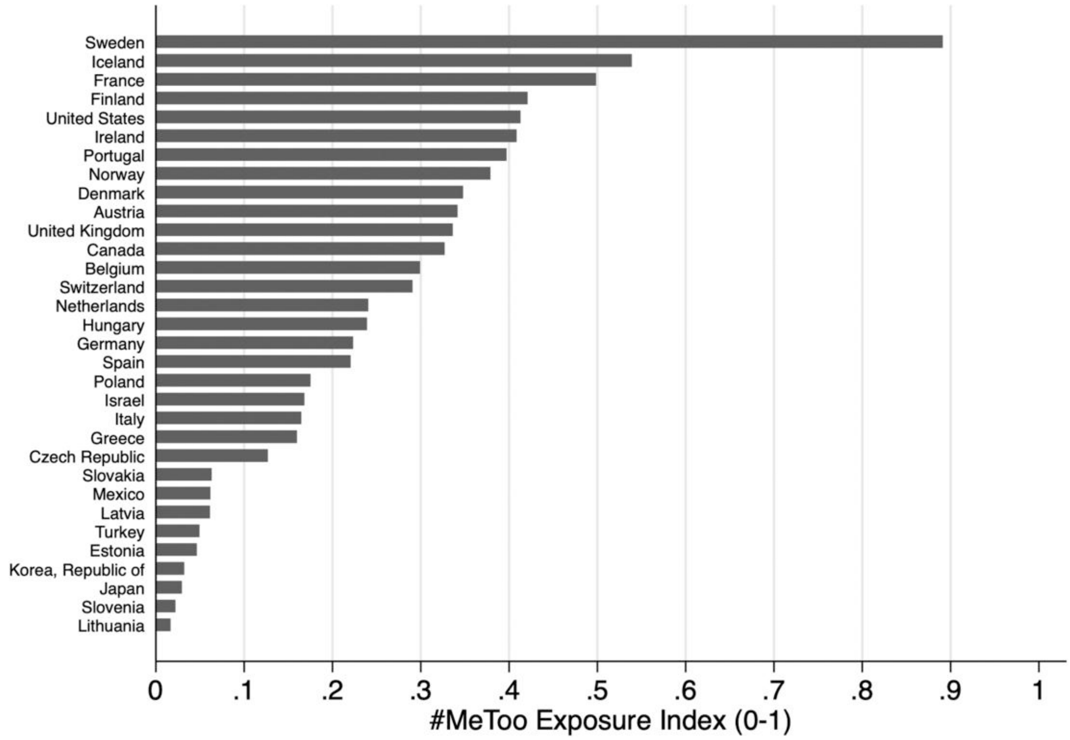
To recover the #MeToo shock β, we rely on the fact that whereas out-of-stock sizes vary between countries in any given week, the same products are proposed in different countries at the same time. In fact, in 90% of cases across the full 2017–2018 panel, the exact same product is available in at least 20 different OECD countries in any given week. This goes up to 98% of cases when we restrict the sample to a window of six weeks before (versus after) #MeToo, with full OECD coverage in 55% of cases and nearly full coverage (i.e., more than 26 countries) in 90% of cases (see Figure S4 in the Online Appendix).

We then exploit the panel structure of the data and differences in the level of #MeToo exposure across countries. Formally, we assume that in countries affected by #MeToo, the market for product i includes the effect of #MeToo, along with a group and time effect:

$$E(S_{i1} | c, t) = E(S_{i0} | c, t) + \beta = \theta_c + \delta_t + \beta,$$

where θ_c is the country-specific effect for countries exposed to #MeToo (c_1) versus unexposed countries (c_0),

Figure 2. #MeToo Exposure Index for Major OECD Countries



Notes. For each country included in the data set, this figure displays a composite index created by averaging the country's scores on three different measures of #MeToo exposure, all of which have been normalized on a 0–1 scale. These measures include (i) the number of #MeToo tweets per thousand users in the six weeks following October 15, 2017, (ii) the count of news headlines related to #MeToo (per million residents) published in major national news outlets during this same six-week period, and (iii) the increase in (normalized) country-level Google search rates focused on sexual harassment-related cases over the same six-week interval.

and δ_t is a time (or seasonal) effect for the pretrend (t_0) and posttrend (t_1). Assuming the time trend δ_t is not country specific, simple difference-in-differences (DD) will identify the effect of #MeToo on exposed countries:

$$DD = [E(S_i|c_1,t_1) - E(S_i|c_1,t_0)] - [E(S_i|c_0,t_1) - E(S_i|c_0,t_0)] = \beta.$$

However, in the presence of country-specific time trends $\delta_{c,t}$ (i.e., sales being on average higher in exposed countries after #MeToo relative to before #MeToo for reasons other than the movement itself), DD leads to

$$DD = \beta + (\delta_{c_1,t_1} - \delta_{c_1,t_0}) - (\delta_{c_0,t_1} - \delta_{c_0,t_0}).$$

This means that the DD estimator will be biased whenever $(\delta_{c_1,t_1} - \delta_{c_1,t_0}) \neq (\delta_{c_0,t_1} - \delta_{c_0,t_0})$. This formalizes the usual common trend assumption, that is, the idea that the sales of shoes in exposed and unexposed countries should exhibit similar patterns in the absence of the #MeToo shock, on average. This condition is unlikely to hold when examining average product sales within countries, but is much more likely to hold when comparing trends for different types of products related to women's identity expression. The main identification strategy hence relies on a three-way interaction between (i) differential exposure to #MeToo across countries, (ii) the timing of the #MeToo shock (pre- versus post-October 15, 2017), and (iii) the presence (versus absence) of stereotypical markers of femininity on a given product.[25]

This specification allows for country-specific time effects $\delta_{c,t}$. Indeed, the identification of the #MeToo effect now relies on a *triple-difference*—or difference-in-difference-in-differences (DDD)—estimator (Blundell and Dias 2009, Olden and Moen 2022), where we compare the number of product stockouts in exposed (c_1) versus unexposed (c_0) countries in the six weeks after (t_1) versus before (t_0) #MeToo began for stereotypical (i_1) versus nonstereotypical (i_0) products. Formally, it can be written as the difference between two difference-in-differences estimators:

$$\begin{aligned} DDD = {} & [E(S_i|c_1,t_1,i_1) - E(S_i|c_1,t_0,i_1)] - [E(S_i|c_0,t_1,i_1) \\ & - E(S_i|c_0,t_0,i_1)] - [E(S_i|c_1,t_1,i_0) - E(S_i|c_1,t_0,i_0)] \\ & - [E(S_i|c_0,t_1,i_0) - E(S_i|c_0,t_0,i_0)] \\ = {} & \beta. \end{aligned}$$

Although the strength of the MeToo movement is not random (e.g., related to media access), and although some countries may be more or less sensitive to the

#MeToo shock (e.g., via customers' sensitivity to gender equality issues), β remains an unbiased estimate of the #MeToo effect as long as #MeToo exposure itself remains uncorrelated with a third factor affecting the market for stereotypical products differently from other products immediately after October 15, 2017. This is a strong condition which would require a sudden change in the weeks following the movement's start that would affect these products differently in exposed countries relative to unexposed countries.

4.2. Estimation Procedure

We test whether #MeToo causally impacted the market for stereotypical products. The main specification exploits within-product-country changes in the average daily number of out-of-stock sizes week to week. We estimate the DDD parameters β from the following fixed-effects Poisson model:

$$\begin{aligned} E[S_{ict}|i,c,t] = exp\{&\beta(Post \times \#MeToo \times Stereotypical)_{ict} \\ &+ \alpha(Post \times \#MeToo)_{ct} \\ &+ \theta(Post \times Stereotypical)_{it} + \gamma Post_t \\ &+ \eta Ln(Price)_{ict} + \Gamma_{ic} + \Omega_{ct}\}, \end{aligned} \tag{1}$$

where S_{ict} reflects the average number of out-of-stock sizes for product i in country c during week t. Because different countries face different customer needs, the retailer may choose different product supply and distribution decisions in each country. We are therefore only focusing on post-#MeToo changes in stockouts for any given product within each country, so Equation (1) includes product-country fixed effects Γ_{ic}.[26]

The main analysis employs a triple-difference (DDD) approach. The parameter β represents the three-way interaction among (i) a binary product-level indicator distinguishing stereotypical products from nonstereotypical products, denoted as *Stereotypical*; (ii) a binary country-level variable indicating exposure to the #MeToo movement, referred to as *#MeToo*; and (iii) a time-specific binary variable distinguishing purchases made during the six weeks following the #MeToo movement from those made in the six weeks preceding it, referred to as *Post*.[27]

The primary analysis focuses on a three-month window, spanning six weeks before and six weeks after the emergence of #MeToo in 2017. This duration accounts for the typical turnaround time in the fashion industry, which can be as short as two weeks. Interpreting any effect as causal beyond this six-week period becomes arduous because retailers may respond to declines in product demand by making adjustments to their product assortments and business strategies. During this three-month period, a total of 411 models of women's shoes were introduced for sale. The majority of these products were available for online purchase throughout the full 12-week period, with an average online presence of approximately 10 weeks per product. On any given day, there was an average of 1.6 out-of-stock sizes per product (with a standard deviation of 2.24). This timeframe also exhibits a well-balanced distribution of observations across countries (see Table S2 in the Online Appendix), with the vast majority of products available throughout the entire OECD (see Figure S4 in the Online Appendix).

The primary specification characterizes a stereotypical product as one that possesses at least two out of three stereotypical markers of femininity: it belongs to the "heels and pumps" category, it is either pink or red, and it features high heels. Out of the 1,718 products available in the women's wear section over the 2017–2018 period, 234 products fall into the "stereotypical" category, which represents about 13.5% of the total.[28] As a measure of a country's exposure to #MeToo, we utilize the #MeToo exposure index discussed in Section 3.3. We create a binary variable by partitioning the continuous index into "unexposed" countries, constituting the bottom tier (11 countries), and "exposed" countries, encompassing the remaining 21 countries. Exposed countries exhibit an average exposure index of 0.35, whereas unexposed countries possess an index close to zero (0.06).[29]

We acknowledge that some countries may have unique local sales periods that influence consumer purchasing patterns. To account for this, we introduce controls for the natural logarithm of the weekly product prices in each country, represented by the parameter η.[30] Furthermore, we address seasonality by including a set of country-month or country-week fixed effects denoted as $\Omega_{c,t}$. It is worth noting that country-week fixed effects do not allow us to separately identify the two-way parameter α, as it gets absorbed by the country-week fixed effects. However, the three-way interaction coefficient from the DDD estimation remains identifiable.

Considering that we are dealing with count data and that the stockout measure includes zero values, we estimate Equation (1) using a Poisson quasi-maximum likelihood model. The Poisson model facilitates the interpretation of the parameters as percentage changes in the number of weekly stockouts. In simpler terms, a negative and significant β implies that, in strongly exposed countries, gender-stereotypical products experienced a β% reduction in product stockouts after #MeToo, relative to nonstereotypical products. We calculate robust standard errors, clustered at the country level. The results remain robust even when alternative clustering of standard errors is applied, as shown in Table S3 in the Online Appendix.[31]

5. Main Results

5.1. Triple-Difference Estimation

Table 1 reports the main effect of the #MeToo movement in exposed countries on the consumption of gender-

Table 1. Triple-Difference Estimation of #MeToo Effect on Gender-Stereotypical Products

	(1)	(2)	(3)	(4)	(5)	(6)
Ln(product price)				−3.075***	−2.957***	−2.735***
				(0.301)	(0.296)	(0.267)
Post	0.186***	0.242***	0.182***	0.122***	0.039	
	(0.040)	(0.023)	(0.042)	(0.036)	(0.045)	
Post × #MeToo	0.055		0.090*	0.098**	0.060	
	(0.045)		(0.049)	(0.042)	(0.050)	
Post × Stereotypical		−0.144***	0.032	0.037	0.024	0.015
		(0.035)	(0.048)	(0.048)	(0.046)	(0.047)
Post × Stereotypical × #MeToo			−0.257***	−0.259***	−0.259***	−0.255***
			(0.057)	(0.058)	(0.054)	(0.055)
Observations	107,912	107,912	107,912	107,912	107,912	107,912
Product × Country FEs	Yes	Yes	Yes	Yes	Yes	Yes
Country × Month FEs	—	—	—	—	Yes	—
Country × Week FEs	—	—	—	—	—	Yes

Notes. Poisson-FE estimates of the influence of #MeToo exposure on weekly stockouts of stereotypical products. We focus on the six weeks following October 15, 2017, compared with the six weeks before. The data comprise 411 products available in 32 OECD countries. In all columns, product-country fixed effects are included. Column 4 adds the log of weekly product prices. Column 5 introduces country-month fixed effects. Column 6 includes country-week fixed effects. Unexposed countries belong to the bottom tier of the #MeToo exposure index. Stereotypical products are defined as those with at least two stereotypical markers of femininity. Robust standard errors clustered at the country level are reported in parentheses.

$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

stereotypical products using the fixed-effects Poisson model presented in Equation (1).

In column 1, we examine the post-#MeToo trends in stockouts for the average product, encompassing both stereotypical and nonstereotypical items. We observe that six weeks after #MeToo, there is an approximately 18.6% increase in the number of out-of-stock sizes. However, this reflects a more general trend that does not significantly differ between exposed and unexposed countries. In column 2, we conduct a similar estimation, this time focusing on post-#MeToo trends in stockouts for nonstereotypical and stereotypical products without distinguishing between exposed and unexposed countries. In this case, we identify a significant 14.4% reduction in product stockouts for stereotypical products relative to nonstereotypical items.

Column 3 presents estimates from the full three-way interaction model. The previously observed reduction in stockouts is primarily concentrated in exposed countries. Specifically, we identify a nearly 26% decrease in stockouts for stereotypical products in exposed countries, which is statistically significant at the 1% level. This finding suggests women consumers responded less to gender-stereotypical products, leading to reduced pressure on available stock. Notably, the difference-in-differences coefficient *Post* × *Stereotypical* is close to zero, suggesting that the decline stems from reduced consumption of stereotypical products in exposed countries rather than an increase in the consumption of stereotypical products in unexposed countries. On the other hand, nonstereotypical products exhibit a weakly significant 9% increase in stockouts in exposed countries (p = 0.065).

In column 4, we introduce controls for the logarithm of product prices to account for potentially localized sales periods around #MeToo. This also enables us to test whether the effect is at least in part driven by the supplier's reaction to #MeToo. The estimation reveals a negative coefficient on the log of product prices of approximately −3. This means that for every 1% increase in product prices, we observe a 3% reduction in product stockouts. It is worth noting that this relatively large estimate aligns with previous estimates regarding the own price elasticity of demand for apparel and footwear, especially in the context of online retail.[32] For instance, studies examining the consumption response to sales tax, such as Agarwal et al. (2017), have reported price elasticities on the order of −10 for apparel. Meanwhile, Einav et al. (2014) found preferred estimates closer to −2.

Taking price changes into consideration reduces the overall upward trend in stockouts from around 18% to approximately 12%. However, the triple-difference estimate remains unchanged whether controlling for product prices or not. Such a pattern mitigates the possibility that the effect would be due to the supplier's reaction to #MeToo.[33] Incorporating country-month fixed effects helps capture most of the remaining seasonal trend in product stockouts post-#MeToo (column 5). In contrast, the triple-difference coefficient remains unaffected. Column 6 introduces our main specification, which includes country-week fixed effects. We estimate a drop in stockouts of approximately 25% for stereotypical products in exposed countries. In sum, and unlike nonstereotypical products, the impact of #MeToo exposure on stereotypical products is both sharp and robust.

5.2. Magnitude

The estimated 25% reduction in product stockouts corresponds to a decrease in the number of out-of-stock sizes for gender-stereotypical products from an average of 2 sizes before #MeToo to 1.5 sizes (out of 8 possible sizes) on average. This decline in pressure on available stock holds economic significance. If we assume that the price parameter estimate from Table 1 accurately reflects the true price elasticity of women's footwear, this effect is akin to imposing an 8% sales tax or price increase on gender-stereotypical products.

To put this effect into perspective, we can also compare it to the impact of "Black Friday." Figure S2 in the Online Appendix illustrates that the average number of out-of-stock sizes increases from 1.5 sizes per product just before Black Friday to slightly above 3 sizes per product, signifying nearly a 100% surge in stockouts. Although the #MeToo effect is concentrated on a smaller subset of stereotypical products, it is roughly equivalent to a quarter of the opposite effect seen during Black Friday.

Additionally, we can gauge the magnitude of the #MeToo effect by considering other #MeToo studies that explore various outcomes related to the movement. For example, Levy and Mattsson (2023) found that during the first six months of #MeToo, there was a 10% increase in the reporting of sexual crimes to the police,[34] whereas Batut et al. (2021) observed a 9.3% increase in the relative share of women leaving firms with a high risk of sexual harassment. Considering that reporting sexual crimes or leaving one's job is a far more consequential decision than making or postponing product purchases, the estimate's magnitude seems plausible.

5.3. Common Trend Assumption

The identification strategy hinges on the assumption that the treated group would have exhibited a similar trend over time as the control group if they had not been exposed to the movement. To validate this common trend assumption, we conduct several tests.

First, we examine high-frequency changes in online search behavior related to the #MeToo movement itself. We compare the evolution of search rates for the sexual harassment topic six weeks before and after October 15, 2017, in exposed countries relative to unexposed countries (see Figure A.2 in the appendix).[35] The analysis confirms the absence of preexisting trends in sexual harassment search rates before #MeToo. Specifically, we observe a significant surge in search rates for sexual harassment cases after October 15, 2017. This surge is predominantly concentrated within the group of exposed countries, with minimal change in unexposed countries. Additionally, we note some nonlinear patterns in how the movement unfolded within those six weeks, with a substantial increase in search rates during the third week of October, followed by a decline, and a gradual rebound with a new peak in search rates during the second week of November. When examining data from 2018, we do not find any significant changes in sexual harassment search rates around October 15.

Next, we assess the presence of consistent pretrends in product stockouts between exposed and unexposed countries around #MeToo. To ensure that both groups demonstrated statistically similar stockout frequencies for products with stereotypical markers relative to other products before #MeToo, we employ the following modified version of Equation (1):

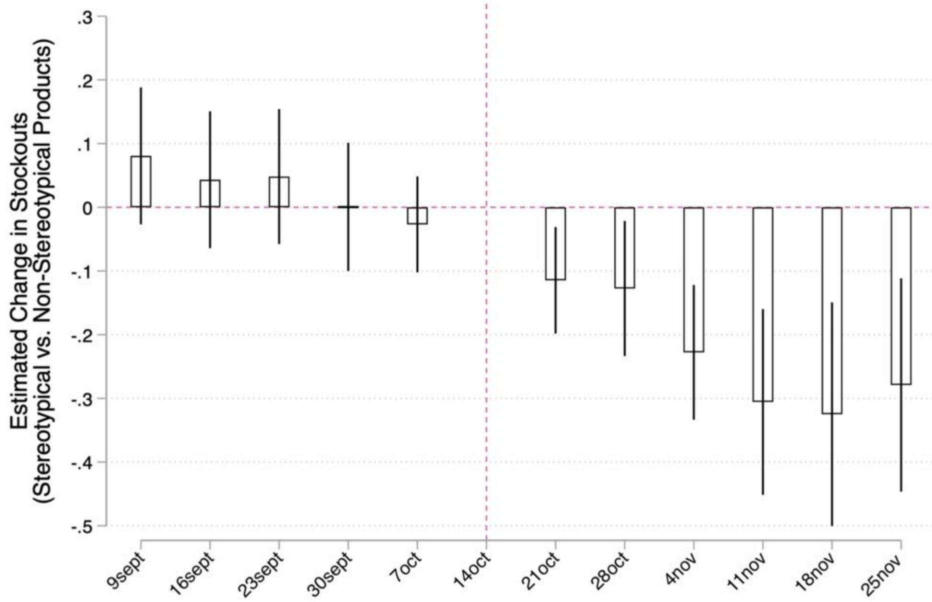
$$E[S_{ic\tau}|i,c,\tau] = exp\left\{\sum_{\tau=-5}^{6}\beta_{\tau}(Post\times\#MeToo\times Stereotypical)_{ic\tau} + \sum_{\tau=-5}^{6}\alpha_{\tau}(Post\times\#MeToo)_{c\tau} + \sum_{\tau=-5}^{6}\theta_{\tau}(Post\times Stereotypical)_{i\tau} + \sum_{\tau=-5}^{6}\gamma_{\tau}Post_{\tau} + \eta Ln(Price)_{ic\tau} + \Gamma_{ic} + \Omega_{c\tau}\right\}, \quad (2)$$

where coefficient β_{τ} captures the three-way interaction DDD parameter on stereotypical products for week τ in comparison with the reference week, in this case the week preceding October 15, 2017 ($\tau = 0$). The common trend assumption implies that coefficients β_{τ} do not significantly differ from zero in the weeks prior to #MeToo (relative to the reference week).

Figure 3 visually demonstrates the common trend assumption by plotting the weekly estimated β_{τ} change in stockouts for gender-stereotypical (versus nonstereotypical) products in exposed (versus unexposed) countries over a 12-week period around October 15, 2017. Before October 15, 2017, there is no notable disparity in stockouts for both types of products between exposed and unexposed countries. However, the first week following October 15, 2017 already displays a sudden and significant drop in out-of-stock sizes. Notably, this drop stabilizes in the second week after October 15, and a more pronounced decline is observed in mid-November, mirroring the pattern seen in the search data related to sexual harassment in Figure A.2 in the appendix. Figure S8 in the Online Appendix replicates the analysis using a wider 24-week window. It confirms that the pretrend fluctuates around zero, whereas the drop stabilizes in the postperiod.

We conduct additional placebo tests by altering the start date of the #MeToo movement and estimating Regression (1) six weeks before and after each fictional starting date. We consider six fictional starting dates: 6 weeks, 9 weeks, 12 weeks, and 15 weeks before and after October 15. The estimated DDD coefficient for each

Figure 3. (Color online) Change in Weekly Stockouts of Gender-Stereotypical (vs. Nonstereotypical) Products Before (vs. After) October 15, 2017, in Exposed (vs. Unexposed) Countries



Notes. Weekly β_τ coefficients estimated using our main specification from Poisson-FE model (2), reported with 95% confidence intervals. Robust standard errors clustered at the country level are in parentheses. The model is estimated over a 12-week period around October 15, 2017, with the week before October 15, 2017 ($\tau = 0$), used as the reference week.

placebo date is plotted in Figure A.3 in the appendix, alongside the baseline estimate with the true starting date. None of the placebo dates yield a triple-difference estimate significantly different from zero.

5.4. Alternative Triple-Difference Estimation

Although we primarily interpret the triple-difference estimate as an average treatment effect on the treated (ATT), it could alternatively be viewed as the difference between two ATTs. Notably, the *Post* × *#MeToo* coefficient loses significance after including country-month fixed effects, but the point estimate remains positive (Table 1, column 5). Instead of interpreting this coefficient as capturing country-specific upward trends $\delta_{c,t}$ in average product stockouts that would have occurred in the absence of a #MeToo shock, we can consider it as a #MeToo effect on nonstereotypical products. This can reflect substitution effects between stereotypical and nonstereotypical purchases. Such substitutions may not necessarily occur if exposed consumers choose to delay their purchase or buy alternative products from other stores.

To provide an alternative perspective, we introduce a modified specification for the triple-difference estimation that aligns more closely with the single ATT interpretation. In this alternative approach, we draw from the sample of men's shoes and create a new categorical variable labeled "*Stereotypical*" that categorizes products into three groups: those from the men's section (*Stereotypical* = 0), which should remain unaffected by the #MeToo movement; those from the women's section but classified as nonstereotypical (*Stereotypical* = 1); and those from the women's section categorized as stereotypical (*Stereotypical* = 2). By designating men's products as the new reference category, we can now estimate two distinct triple-difference coefficients, one for stereotypical and one for nonstereotypical women's products.

We first replicate the main results using the women's sample exclusively, with the post-#MeToo exposure effect estimated separately for stereotypical and nonstereotypical products (Table A.1, panel A, in the appendix). These estimates can be compared with the triple-difference estimates calculated using men's products as the reference category. Although exposed countries do indeed experience higher product stockouts post-#MeToo, this trend is consistent with a general seasonal pattern in footwear sales that also affects men's products (Table A.1, panel B, in the appendix). In relative terms, there is no significant change in stockouts for nonstereotypical women's products compared with men's products, but there is a substantial and statistically significant 22% decrease in stockouts for stereotypical women's products. In other words, the #MeToo effect seems primarily driven by the weaker adherence to the most stereotypically feminine items, rather than by an elevated adherence to more gender-neutral or masculine products.

5.5. Other Robustness Tests

5.5.1. Control Group. The main specification assumes that countries ranked below the bottom tier of exposure were not significantly affected by #MeToo. However, a few of these countries may have experienced some level of impact, suggesting that the estimated 25% drop could underestimate the true effect. To investigate this, we use the bottom quartile of exposure instead of the bottom tier to define unexposed countries, and this adjustment does not substantially alter the results (Table 2, column 2). Conversely, if we choose a higher threshold for defining exposed countries, it may lead to a lower estimated treatment effect as some countries that were previously exposed are now included in the control group. In this analysis, we follow Levy and Mattsson (2023) and distinguish between a "strong" (above median exposure) and a "weak" (below median exposure) #MeToo movement. The estimated effect on the triple difference decreases from an estimated 25% drop to a 17% drop, but remains highly significant (Table 2, column 3). In column 4, we assess the effect of a strong (versus weak) #MeToo movement within the group of exposed countries. Unexposed countries (bottom quartile) are removed from the sample, and we define strongly exposed countries as those ranking above the median in exposure within the group of 24 exposed countries. This results in a decrease in the estimated effect on the triple difference from 25% to about 14%, but it remains significant. Finally, we use the #MeToo exposure index as a continuous measure of treatment intensity in column 5.[36] Each one-standard-deviation increase in exposure is associated with a nearly 11% reduction in stockouts for stereotypical products (Table 2, column 5).

5.5.2. #MeToo Exposure Measurement. The primary analysis relies on the composite index as a comprehensive measure of #MeToo exposure. However, we demonstrate the robustness of the results by replicating them using each of the three separate measures of #MeToo exposure (Table 2, columns 6–8). First, regardless of the measure used, the triple-difference estimates are consistently negative and fall within the same confidence interval, demonstrating the consistency of the findings.[37] Second, the *Post* × *Stereotypical* coefficients, which capture the prepost change in stockouts of stereotypical products in unexposed countries, are all close to zero and nonsignificant. This confirms the effect is driven by a change in stockouts within the exposed countries rather than by the reverse change within the control group.

5.5.3. Stereotype Measurement. The main specification defines a stereotypical product as having at least two markers of femininity, classifying about 11% of products as stereotypical. As a less conservative approach, we can define stereotypical products as having at least one marker of femininity, resulting in approximately 49.5% of products classified as stereotypical. Although the effect remains highly significant, the estimated #MeToo effect is nearly halved, moving from −25% to about −14% (Table 2, column 9). This implies that the majority of the reduction in product stockouts is driven by the most highly stereotypical items. To further enhance the robustness of the analysis, we assign an SI to each shoe model in the retailer data set.[38] We then create new binary variables that define a stereotypical product as having a predicted SI greater than or equal to six (i.e., "feminine" or "very feminine") or equal to seven ("very feminine").[39] Using these two measures, we estimate a 13% drop (or a 32% drop) in product stockouts, which are of comparable magnitude as using one (or two) stereotypical markers as the definition of stereotypical products (Table 2, columns 10–11). We also employ the stereotypical index as a continuous measure (z-scored) in column 12. The triple-difference coefficient is negative and highly significant, indicating that, relative to the average product, products ranking two standard deviations higher in SI experienced a 17% drop in stockouts. Finally, we estimate difference-in-differences coefficients for each shoe color and plot the coefficients against their respective SI.[40] Figure S10 in the Online Appendix shows a clear decline in stockouts for more stereotypically feminine colors in exposed countries after #MeToo.

5.5.4. Time Window. The #MeToo movement officially began on October 15, 2017, and our preferred specification uses a 12-week window (6 weeks before and after the start of the movement) to estimate its impact. We favor this timeframe because it can be challenging to interpret longer-term estimates as causal. Over a more extended period, retailers might adjust their strategies, potentially replacing stereotypical products with less stereotypical alternatives. Furthermore, the control group may no longer be valid if some initially unexposed countries become exposed in the long run. For example, Japan's #KuToo movement, advocating against workplace dress codes requiring women to wear high heels, emerged in 2019 as a follow-up to #MeToo. The choice of a different time window could also impact the estimate if the effect significantly increases or diminishes over time. As observed in Figure 3, the effect is sudden, manifesting itself in the first week after #MeToo, and becomes stronger in the fourth week. To assess the sensitivity of the results to the time window, we vary the estimation period from 6 weeks before and after to 3 weeks, 9 weeks, 12 weeks, and 15 weeks before and after, while keeping the true starting date constant. Using a shorter 3-week window reduces the point estimate from −25% to −15%, whereas choosing a somewhat larger time window slightly increases the estimate to −30% (see Table S4 in the Online Appendix).[41]

Table 2. Robustness

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Post × Stereotypical	0.015	0.030	−0.073	−0.139***	−0.156***	−0.066	−0.020	−0.013	0.046	−0.047	−0.019	−0.038*
	(0.047)	(0.051)	(0.047)	(0.046)	(0.027)	(0.071)	(0.058)	(0.055)	(0.033)	(0.032)	(0.082)	(0.019)
Post × Stereotypical × #MeToo	−0.255***	−0.243***	−0.171***	−0.144**	−0.106***	−0.140*	−0.209***	−0.212***	−0.137***	−0.134***	−0.323***	−0.085***
	(0.055)	(0.060)	(0.060)	(0.056)	(0.040)	(0.077)	(0.066)	(0.064)	(0.048)	(0.050)	(0.090)	(0.029)
Observations	107,912	107,912	107,912	82,597	107,912	107,912	107,912	107,912	107,912	107,912	107,912	107,912
#MeToo measurement												
Composite index	Yes	Yes	Yes	Yes	Yes	—	—	—	Yes	Yes	Yes	Yes
Tweets per million users	—	—	—	—	—	Yes	—	—	—	—	—	—
Number of headlines	—	—	—	—	—	—	Yes	—	—	—	—	—
Sex. harassment search	—	—	—	—	—	—	—	Yes	—	—	—	—
Control group	<p33	<p25	<p50	<p50	—	<p33	<p33	<p33	<p33	<p33	<p33	<p33
Continuous (z-score)	—	—	—	—	Yes	—	—	—	—	—	—	—
Stereotype measurement												
Stereotypical markers	2+	2+	2+	2+	2+	2+	2+	2+	1+	—	—	—
Stereotypical index ≥ 6	—	—	—	—	—	—	—	—	—	Yes	—	—
Stereotypical index = 7	—	—	—	—	—	—	—	—	—	—	Yes	—
Stereotypical index (z-score)	—	—	—	—	—	—	—	—	—	—	—	Yes
Sample												
OECD countries	All (32)	All (32)	All (32)	Treat (24)	All (32)	All (32)	All (32)	All (32)	All (32)	All (32)	All (32)	All (32)
Fixed effects												
Product × Country FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country × Week FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Poisson-FE estimates of the influence of #MeToo exposure on weekly stockouts of stereotypical products. All models control for the log of weekly product prices and include product-country and country-week fixed effects. Column 1 is the main specification. Column 2 defines the control group of unexposed countries as belonging to the bottom quartile of exposure, and column 3 uses weakly exposed countries (below median exposure). Column 4 defines weakly exposed countries as below median exposure within the group of exposed countries only. Column 5 uses the #MeToo exposure index as a continuous (z-scored) measure of treatment intensity. Columns 6, 7, and 8 replace the index by the number of #MeToo tweets per million users, the number of #MeToo headlines per million residents, and the increase in sexual harassment-related searches, respectively. Column 9 classifies stereotypical products as having at least one marker of stereotypical femininity, column 10 as having a (rounded) predicted stereotypical index higher than or equal to six ("feminine" or "very feminine"), and column 11 as equal to seven ("very feminine"). Column 12 uses the stereotypical index as a continuous (z-scored) measure. Robust standard errors clustered at the country level are in parentheses.

$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

5.5.5. Ordinary Least Squares–Fixed-Effects (OLS-FE) Estimates. Table S5 in the Online Appendix presents estimates using OLS fixed effects and alternative transformations of the dependent variable. First, we apply the inverse hyperbolic sine (IHS) transformation to the stockout variable, which approximates a log transformation when zero values are present. The OLS-FE triple-difference estimate reveals a nearly −17% effect size (see Table S5, column 2, in the Online Appendix). The coefficient on unexposed countries remains close to zero. Next, we create two alternative measures of stockouts: one capturing the fractional share of out-of-stock sizes per product and another represented by a binary dummy variable indicating whether a particular product experienced at least one out-of-stock size in a given week. The linear probability model yields highly consistent estimates. The #MeToo effect results in a 5.1-percentage-point reduction in the stockout rate of stereotypical products, equivalent to a 22% drop relative to their pre-#MeToo baseline (see Table S5, column 3, in the Online Appendix). Stereotypical products are also 8 percentage points less likely to have at least one size out of stock (see Table S5, column 4, in the Online Appendix).

5.5.6. Daily Data. Our preferred specification aggregates product stockouts at the weekly level because of noise reduction and computational efficiency. However, we also explore the robustness of the results using daily stockout data, which increases the sample size by a factor of seven. Table S6 in the Online Appendix confirms the consistency of the findings using daily data. Additionally, we estimate a linear probability model at the country-product-size level using OLS. This approach considers each size as either out of stock (zero) or in stock (one) for a given product. Given that each product has exactly eight possible sizes, the sample reaches a total of about 6.7 million daily observations. We include product-size-country fixed effects so that each product size is considered as a single product. On average, relative to nonstereotypical products, a particular size of a stereotypical product is 5.2 percentage points less likely to be out of stock, corresponding to a nearly 27% reduction relative to the pre-#MeToo baseline (see Table S5, column 5, in the Online Appendix).

5.5.7. Seasonal Trends. We perform additional placebo tests to rule out the possibility of omitted seasonal trends specific to exposed countries affecting the market differently for more colorful or bulkier products exclusively in the postperiod. We first ask whether the effect also occurs within the men's section of the website. We replicate the triple-difference estimation within the men's sample using the 250 men's products sold during the same 12-week window. Because men's shoes lack the same distribution of feminine, neutral, and masculine markers as women's shoes, we cannot use the same binary measure for stereotypically feminine items. As an alternative, we create a placebo stereotypical index for the men's shoe sample, using the same estimates derived from the analysis of women's product pictures.[42] We investigate whether a one-standard-deviation increase in the index within the men's shoe sample (e.g., because of pink or red shoes) exhibits a similar reduction in stockouts as what we find for women's shoes. Contrary to women's products, we find no significant effect on the three-way interaction for men's products (see Table S7, column 1, in the Online Appendix). The observed effect is hence not reflective of a broader decline in the popularity of colorful shoes compared with black sneakers, as such trends would also manifest within the men's section. Second, we replicate the analysis using a 12-week window around October 15, 2018.[43] We analyze the 427 products sold six weeks before and after October 15, 2018. Using 2018 instead of 2017, we find no effect around the same period, indicating that general seasonal effects replicating year to year do not drive the observed effect (see Table S7, column 2, in the Online Appendix).[44] Finally, we add product-country linear time trends based on the preperiod to our main specification. The point estimate is slightly reduced but remains within the same confidence interval (see Table S7, column 4, in the Online Appendix).

5.6. Generalizability to Other Product Categories

We chose to focus on a product category where gender associations range from very feminine to somewhat feminine, more neutral, or even somewhat masculine. For instance, products like women's sneakers and men's sneakers often share similar designs, materials, and color palettes. As a result, gender stereotyping in these categories is also more salient. Conversely, exclusively feminine product categories such as bras or dresses do not share this key feature of the empirical setting. These products carry deep-seated cultural and historical connotations that reinforce their gender associations, rendering them less adaptable to gender-neutral or masculine interpretations.

Rather than an increased adherence to more neutral (or masculine) items, the findings suggest that the #MeToo effect stems from reduced adherence to the most stereotypically feminine items within footwear. Consequently, we may still be able to identify similar effects within product categories that are already perceived as feminine, especially if we focus on the most stereotypically feminine items. To expand the analysis, we hence collected stockout and pricing data for three additional women's product categories: lingerie products (55 products), dresses (985 products), and handbags (444 products). Although we lack survey-based measures of stereotypical femininity for these products, we used the color pink or red as a clear and widely applicable indicator of stereotypical femininity, ensuring a consistent basis for comparing effect sizes across all four product categories.

The analysis reveals consistent reductions in stockouts for pink or red items across these product categories, albeit with varying degrees of magnitude and precision in estimates. These reductions range from 6.5% for pink or red dresses to an 18% decrease for handbags and a 22% reduction for shoes, and the most important (although weakly significant) decrease is observed in pink or red lingerie products, with a nearly 50% decline (Table 3). We also look for pretrend in pink or red product stockouts between exposed and unexposed countries in the 12 weeks before (versus after) #MeToo (see Figure S9 in the Online Appendix). Although the parallel trend is less clear in this case, the β_τ coefficients are either positive or not significantly different from zero prior to #MeToo, contrasting with being negative or not significant after #MeToo.

6. Channels and Potential Behavioral Mechanisms

6.1. Supply-Side Effects

The #MeToo movement led to a significant decline in the market for stereotypical products. This decline can be attributed to two primary channels: a swift shift in consumer preferences, which, in turn, decreased pressure on existing inventory, or alternatively, retailers embracing and adapting to their customers' reactions by altering their strategy. Specifically, retailers may proactively foresee diminished demand for stereotypical products in regions where customers have been exposed to the movement, prompting them to adjust their product offerings and pricing strategies accordingly.

We show that changes in product assortments cannot explain the documented effect. Firstly, the dependent variable captures changes in out-of-stock sizes *within* country-product, focusing on products available on the platform for almost the entire 12-week window (average of 11.1 weeks per product). Changes in assortments are rare within this relatively short time window, and unlikely to drive the effect. This is supported by Table S8 in the Online Appendix, which tests the robustness of the main effect when restricting the sample to products that were never removed (column 2), were never added (column 3), or were neither added nor removed (column 4) throughout the entire 12-week period. Moreover, the results remain consistent whether selecting products based on their availability in all OECD countries or selecting them based on availability only in some but not all OECD countries (see Table S9 in the Online Appendix).[45]

Secondly, we examine the impact of exposure to #MeToo on the likelihood of a particular product being added or removed over the 12-week period. We create a binary variable for each product to capture whether it was added (equal to one in the week it was added and subsequent weeks, and zero otherwise) or removed (equal to one in the week it was removed and subsequent weeks, and zero otherwise). We conduct a triple-difference analysis to assess whether stereotypical products were more or less likely to be removed or added after #MeToo in countries exposed to the movement. The results, based on an OLS-FE linear probability model, reveal that although stereotypical products were indeed more likely to be removed and less likely to be added after #MeToo, this pattern is observed in both unexposed and exposed countries (Table 4, columns 1–4).

Table 3. Footwear vs. Other Product Categories

	Shoes		Lingerie		Dresses		Handbags	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post	0.132***		0.307***		0.162***		0.023	
	(0.035)		(0.045)		(0.018)		(0.061)	
Post × #MeToo	0.089**		−0.036		0.030		0.110	
	(0.041)		(0.057)		(0.022)		(0.067)	
Post × Pink/Red	−0.048	−0.059*	−0.205	−0.262	−0.002	−0.013	−0.049	−0.048
	(0.035)	(0.031)	(0.253)	(0.232)	(0.030)	(0.029)	(0.091)	(0.103)
Post × Pink/Red × #MeToo	−0.235***	−0.225***	−0.556*	−0.486*	−0.065**	−0.064**	−0.199**	−0.183*
	(0.048)	(0.045)	(0.286)	(0.279)	(0.031)	(0.029)	(0.099)	(0.109)
Observations	107,912	107,912	5,235	5,235	175,915	175,915	43,879	43,879
Product × Country FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country × Week FEs	—	Yes	—	Yes	—	Yes	—	Yes

Notes. Poisson-FE estimates of the influence of #MeToo exposure on daily stockouts of stereotypical products across four product categories. We focus on the six weeks following October 15, 2017, compared with the six weeks before. Over this period, the data comprise women's footwear (411 products), lingerie (55 products), dresses (985 products), and handbags (444 products) available in 32 OECD countries. All models control for the log of product prices and include product-country fixed effects. Columns 2, 4, 6, and 8 include country-week fixed effects. Robust standard errors clustered at the country level are in parentheses.

*$p < 0.10$; **$p < 0.05$; ***$p < 0.01$.

Table 4. Supply-Side Effects

	Product removed		Product added		Ln(product price)	
	(1)	(2)	(3)	(4)	(5)	(6)
Post	0.019***		0.210***		−0.025***	
	(0.003)		(0.005)		(0.004)	
Post × #MeToo	0.001		−0.007		0.003	
	(0.004)		(0.005)		(0.004)	
Post × Stereotypical	0.043***	0.043***	−0.133***	−0.134***	−0.003	−0.001
	(0.009)	(0.009)	(0.005)	(0.005)	(0.003)	(0.003)
Post × Stereotypical × #MeToo	−0.006	−0.006	0.010	0.010	−0.004	−0.004
	(0.010)	(0.010)	(0.006)	(0.006)	(0.003)	(0.003)
Observations	146,856	146,856	146,856	146,856	120,149	120,149
Product × Country FEs	Yes	Yes	Yes	Yes	Yes	Yes
Country × Week FEs	—	Yes	—	Yes	—	Yes

Notes. OLS-FE estimates of the influence of #MeToo exposure on a weekly dummy for whether a product has been removed (columns 1–2) or added (columns 3–4) to a country's product assortments. Columns 5–6 use the log of weekly product prices as dependent variable. The data comprise 411 products sold in 32 OECD countries six weeks before and six weeks after the #MeToo shock of October 15, 2017. All models include product-country fixed effects. Columns 2, 4, and 6 add country-week fixed effects. Robust standard errors clustered at the country level are in parentheses.

*$p < 0.10$; **$p < 0.05$; ***$p < 0.01$.

Whereas product assortment decisions may take time to process, pricing strategies such as targeted sales or price adjustments on substituted products can offer a more prompt response to anticipated declines in demand. We therefore also look at whether the price of nonstereotypical and stereotypical products was in any way affected after #MeToo. We find that the post-#MeToo period is associated with a 2.5% reduction in average product prices. Notably, this effect is seasonal and resembles a retailer-level promotion, as it impacts all products and countries, regardless of their exposure to the movement (Table 4, columns 5–6). This may also suggest that the retailer was already pricing its products close to marginal cost before #MeToo, making any downward price adjustment difficult.

Even in cases where the retailer does not change its product assortment or prices, it could still shift quantities between different geographical locations or from local central warehouses among available products. For example, the retailer may reduce the shipment of pink shoes to exposed countries. It could also employ strategies to make gender-stereotypical products less visible on its website or decrease marketing activities related to such products (a form of "cloaking" strategy (Yoganarasimhan 2012)).

Although we cannot fully rule out these possibilities, they alone are unlikely to explain the results. The impact on out-of-stock sizes is both sudden and significant, observable during the first week after October 15, 2017 (as shown in Figure 3). Decisions related to product shipments or changes in advertising budgets typically take time to process, generally extending over a period of two weeks or more, and are often coordinated with fashion seasons. Although it is conceivable that shifts in gender attitudes over previous years or decades contributed to #MeToo, the movement itself could not have been anticipated only a week or two weeks prior, making a supply-side explanation unlikely, at least in the short term.

The evidence presented in the following section, which includes daily variations in both product stockouts and exposure, brings further support to a demand-driven explanation.

6.2. Social Media Activity and Online Search Behavior

Social media activity and online behavior may serve as direct channels through which a digitally native social movement like #MeToo influenced consumer behaviors. We investigate whether the observed effect aligns with the timing of daily tweets related to #MeToo on social media, and the subsequent increase in online searches for sexual harassment cases.

We proceed in two steps. First, we classify products into either a stereotypical group or a nonstereotypical group, using the same definition employed in the primary analysis. Second, we aggregate the data at the product group × country × daily level and estimate a fixed-effect Poisson model on the resulting country-level (balanced) daily panels. This model allows us to predict the average number of product stockouts, denoted as S_{gct}, for a given product group g on day t in country c. This is achieved through the following Poisson quasi-maximum likelihood model:

$$\begin{aligned} E[S_{gct}|g,c,t] = exp\{&\alpha Tweets_{ct} + \beta(Tweets \times Stereotypical)_{gct} \\ &+ \delta NumProd_{gct} + \omega(NumProd \times Stereotypical)_{gct} \\ &+ CountrySeason_{ct} + ProdSeason_{gt} + \nu_{gc} + \Gamma_t\}, \end{aligned} \tag{3}$$

where $Tweets_{ct}$ represents the total number of #MeToo tweets in country c during day t, with the IHS transformation applied to account for zero values. $Stereotypical_{gct}$ is the stereotypical products dummy. To capture product assortment effects, we control for the total daily number of available products within each group, $NumProd_{gct}$. We also include a full set of time dummies Γ_t for date fixed effects and country-product group dummies ν_{gc}, so that any effect on product group stockouts exclusively arises from within-country variation in daily tweets. Finally, we control for (weekly) seasonal effects within country ($CountrySeason_{ct}$) and product groups ($ProdSeason_{gt}$). Robust standard errors are further adjusted for two-way clustering on both countries and days.

Within the post-#MeToo period for which we collected daily tweets about #MeToo (October 2017–November 2018), we find that the semielasticity parameter α is positive, whereas β is negative for stereotypical products (panel A of Table 5, column 1). On days with high #MeToo-related Twitter activity, we observe that stockouts are more common for nonstereotypical products, and significantly less common for stereotypical products. However, a causal interpretation of α and β requires that, given the comprehensive set of fixed effects, any residual variation in #MeToo tweets can be treated as if it were randomly assigned. This may not be the case, as Twitter activity can be influenced by unmeasured third factors that are not accounted for in the analysis. To further explore this relationship and test its robustness, we conduct a Granger test, examining whether #MeToo tweets published on the day before significantly explain changes in product stockouts on the following day. Although the Granger test does not solve the omitted variable bias, it helps in addressing autocorrelation and identifying directions of influence. The estimated association is smaller, but remains statistically significant (panel A of Table 5, column 2).

The impact of #MeToo on social media was immediate and far-reaching, empowering women to share their stories of harassment and finally addressing an issue

Table 5. Daily Stockouts and Online Activity After #MeToo (Panel Analysis)

	(1) Post-#MeToo	(2) Post-#MeToo	(3) Pre-#MeToo
Panel A. Twitter daily panel (2017–2018)			
IHS(Tweets)	0.014***		
	(0.005)		
IHS(Tweets) × Stereotypical	−0.024***		
	(0.007)		
IHS(Tweets) ($t-1$)		0.001	
		(0.003)	
IHS(Tweets) ($t-1$) × Stereotypical		−0.007***	
		(0.002)	
Observations	25,536	25,536	—
Panel B. Google search daily panel (2017–2018)			
Sexual harassment	0.000		−0.002
	(0.006)		(0.005)
Sexual harassment × Stereotypical	−0.017***		−0.001
	(0.005)		(0.009)
Sexual harassment ($t-1$)		0.002	
		(0.001)	
Sexual harassment ($t-1$) × Stereotypical		−0.005***	
		(0.002)	
Observations	27,762	27,697	18,016
Date FEs	Yes	Yes	Yes
Country × Prod. Group FEs	Yes	Yes	Yes
Country × Week FEs	Yes	Yes	Yes
Prod. Group × Week FEs	Yes	Yes	Yes
Product assortment controls	Yes	Yes	Yes
Lagged Dep. Var.	—	Yes	—

Notes. Panel regressions of daily product stockouts on daily tweets (panel A) and Google searches (panel B) estimated using Poisson-FE model. The unit of observation is at the country × product group × daily level. Columns 1 and 2 are estimated on the post-#MeToo period. Column 3 of panel B estimates the same model on the pre-#MeToo period. All regressions include date fixed effects, country-group fixed effects, the log of total daily number of available products within each product group (interacted with product group dummies), and control for country and product group seasonal effects at the weekly level. Column 2 further controls for the lag of the dependent variable (interacted with product group dummies). Robust standard errors are in parentheses, adjusted for two-way clustering on countries and days. Prod., product; Dep. Var., dependent variable.

*$p < 0.10$; **$p < 0.05$; ***$p < 0.01$.

that had long been suppressed. There is a robust positive correlation between daily #MeToo tweets and daily searches related to sexual harassment within countries, as illustrated in Figure S11 in the Online Appendix. Although cases of sexual harassment had been publicly discussed before the advent of #MeToo, the movement led to a substantial surge in daily search volumes on this subject. On average across the 32 countries, searches regarding sexual harassment were more than one standard deviation higher six weeks after #MeToo's start compared with the six weeks before, and this heightened interest persisted throughout 2018 (see Figure S12 in the Online Appendix). In contrast, searches related to "gender equality" did not exhibit dramatic changes after #MeToo, whereas they did experience an increase surrounding the 2017 International Women's Day. It is noteworthy that #MeToo triggered enduring shifts, whereas the impacts of the 2017 and 2018 International Women's Days on Google searches were more transient.

Panel B of Table 5 replicates the previous analysis, but this time, we shift our focus to examine the relationship between sexual harassment-related Google searches (normalized using z-scores) and stockouts, rather than tweets. In column 1, we observe a consistent pattern of reduced stockouts for stereotypical products on days with increased online searches related to sexual harassment, and this finding holds up under Granger testing in column 2. Unlike #MeToo tweets, searches regarding sexual harassment were already occurring before the onset of #MeToo. This allows us to explore whether the pattern identified in column 1 extends to the months leading up to the #MeToo movement. However, as column 3 reveals, we do not find any significant effects when examining daily searches on sexual harassment and daily stockouts before October 15, 2017. These results suggest that #MeToo did more than just raise awareness of sexual harassment cases; it may have ignited a deeper connection between women's disclosures of their harassment experiences and a broader shift away from gender stereotyping. Given the relatively short time frame considered when using tweets, Google searches, and daily stockout changes, these effects appear to reflect shifts in consumer attitudes rather than supply-side factors. The following section will delve into potential behavioral mechanisms that could explain this demand-side effect.

6.3. Discussion on Potential Mechanisms

Although our primary research objective is to provide empirical support for the significant influence of #MeToo on consumption, this section explores three overarching behavioral mechanisms that may account for a shift in demand, serving as potential areas for further investigation.

To inform the discussion, we present additional tests and the outcomes of an online experiment involving about 1,000 women, representative of the U.S. population. In the experiment, respondents were randomly allocated to a control condition, or to one of three treatment conditions, namely, (i) a #MeToo condition in which they were asked to discuss their feelings, thoughts, and engagement with #MeToo; (ii) a threat condition where they described a time they felt threatened as women; and (iii) an empowerment condition where participants shared experiences where they felt empowered. Control respondents were asked to write about a trip to the grocery store. Detailed information about the experiment and treatments can be found in Online Appendix E.

Firstly, it is plausible that #MeToo raised awareness regarding identity-related threats. These threats may manifest in tangible ways, such as fears of harassment and personal safety risks, or in subtler forms like concerns about social stigma, such as "victim blaming." In simpler terms, women might have felt less secure and more anxious about wearing gender-stereotypical products in public immediately after the emergence of #MeToo, especially if these products were seen as attracting unwanted male attention. This hypothesis implies that gender-stereotypical products are more likely to draw men's attention. However, when we measure the impact using a "male attention" scale rather than a gender-stereotypical scale, we find much weaker effects of #MeToo (see Table S10 in the Online Appendix).[46] Moreover, we found a (weakly) significant effect on private consumption (e.g., pink lingerie) that does not align with the effect being only driven by the fear of public shaming (see Table 3). Notably, the results of the online experiment also do not align with the identity threat hypothesis. Although the #MeToo treatment reduces stated preferences for stereotypical attributes, the "threat condition" has no significant impact (see Table S11 in the Online Appendix). Given that both treatments increase sensitivity to threat and decrease sense of power, the significant #MeToo treatment effect likely originates from a distinct, unrelated channel.

A second potential explanation lies in the empowerment of women. #MeToo, with its focus on gender equality at work, women's empowerment, and destigmatizing narratives, empowered women's identities. Social activist Tarana Burke originally used the term "Me Too" in 2006 to promote women's "empowerment through empathy." If stereotypical markers of femininity primarily cater to men's preferences rather than women's comfort and needs, women who felt empowered after #MeToo might have opted to purchase more nonstereotypical products. For instance, women might have chosen more comfortable footwear, like flat shoes, signaling a shift toward valuing their own comfort over adhering to traditional gender norms.[47] However, rather than a fall in the demand for gender-stereotypical products, we would then expect the movement to generate an immediate surge in demand for nonstereotypical items, which is not the case. Additionally, the results of the online experiment do not align with the empowerment hypothesis, as the empowerment treatment condition does not affect preferences for stereotypical footwear (see Table S11 in the Online Appendix).[48] It's worth noting, however, that our focus was on the immediate impact of the movement whereas women's empowerment might be a longer-term consequence of the movement.

Unlike the threat and empowerment conditions, the #MeToo condition does replicate the #MeToo effect in the laboratory experiment. This suggests the effect may be more specific to the movement itself, rather than mediated by identity threat or empowerment. Although the movement primarily focused on voicing harassment stories, it also shed light on how deeply ingrained beliefs about gender roles can contribute to a culture of silence and sexism that enables and perpetuates such behaviors. #MeToo, through its vocalization of harassment cases, challenged the fundamental aspects of patriarchal societies and called for a transformation in prevailing gender roles. The immediate drop in demand for stereotypical markers of femininity may thus symbolize a collective desire to endorse the movement's goals of challenging stereotypical notions of femininity, or to signal a departure from patriarchal perspectives on the subject, whether stemming from internal beliefs or concerns about external perceptions.

To gain further insight into the essence of the #MeToo movement, we draw from the literature on social movements and emotions (Jasper 2011) and analyze the sentiments expressed in women's written narratives. Our focus centers on four core emotions: happiness, sadness, fear, and anger. According to the identity threat explanation, we would expect individuals to primarily express fear when reminded of the movement. In contrast, if empowerment is the driving force, positive emotions like happiness should outweigh negative ones like fear or sadness. Lastly, an explanation grounded in value alignment and rebellion against stereotypes would arguably position anger as the dominant emotion (Van Zomeren et al. 2004). We follow recent advances in sentiment analysis (Rathje et al. 2024) and use ChatGPT to analyze the text written by respondents in each experimental condition.[49] The analysis of emotions confirms the distinctive nature of the #MeToo movement. Fear, although dominant in the threat condition (64% of narratives), accounts for only 4.5% of responses in #MeToo narratives. In contrast, happiness emerges as the primary emotion in the empowerment condition, comprising nearly 63% of narratives (in contrast to a mere 8.5% for sadness). However, within the #MeToo context, happiness stands at roughly 28%, equivalent to the prevalence of sadness. Notably, the #MeToo condition distinguishes itself through the prominent occurrence of anger, representing approximately 23% of narratives, in contrast to the other two conditions, where it represents only 10%.

In conclusion, although we cannot definitively determine how specific emotional forces ultimately influenced behavior, the analysis of women's narratives reveals the distinctive nature of the #MeToo movement. It illustrates a complex emotional landscape where happiness, sadness, and, notably, anger all played significant roles. Rather than being a matter of pure threat or empowerment, the diverse emotional spectrum and the importance of anger align with the political nature of a movement that ignited a profound determination to challenge prevailing gender norms.

7. Conclusion

This paper presents causal evidence of the global impact of the #MeToo movement on consumption behaviors of stereotypical products across major countries. The findings shed new light on how identity-based social movements can lower the consumption of products associated with identity stereotypes. The observed effect is primarily consistent with a demand-side mechanism, and aligns with the temporal patterns of social media activity and online searches generated by the movement.

Our research has broad managerial implications as we show that social movements can impact firms—even when they are not directly targeted—by triggering shifts in consumer preferences. In this context, brand-led aesthetic and stylistic decisions that shape the positioning and offerings of brands become pivotal. Such decisions expose brands not only to changing fashion trends but also to broader shifts related to consumer identities. Our findings can inform the consequences of assortment choices made by firms. These choices position producers in the eyes of consumers, competitors, and other critical audiences. Although firms can develop a market when their assortments align with consumers' self-perceived identities, their market size can shrink when these identities are evolving or less likely to be expressed.

The decision of whether firms should respond by shifting their assortments toward less stereotypical alternatives depends at least in part on the duration of the impact of social movements. Actively monitoring changes in discourses or societal views associated with consumer identities thus becomes increasingly critical for firms that primarily create value through identity-relevant products. In the case of #MeToo, although interpreting long-term changes as causal is difficult, the fact that we do not see a reversal of the effect even after three months suggests that firms not internalizing the effect may suffer from a persistent mismatch between customer demand and their offerings. Even if the effect is short-lived, it may be strong enough for firms favoring less stereotypical items in their product assortments to benefit from positive first-mover advantages. Beyond shifts in production or retail assortment, firms may also decide to become vocal and take a stance. Such actions may, in turn, significantly contribute to building their cultural capital in the long run.

Nonetheless, the current research is not without its limitations, which also offer promising avenues for future work. Firstly, although the evidence suggests a demand-driven effect, the behavioral mechanisms at play remain elusive, largely because of the absence of consumer-level information. Besides eliciting customer-level beliefs directly, for instance, through surveys, one promising direction for future research is to delve into the narratives and sentiments expressed in social media as a means to better understand these mechanisms. For instance, the #KuToo movement in Japan, which challenges workplace policies mandating women to wear high heels, has drawn parallels between this practice and female foot binding, emphasizing the normative influence of activists. In different countries, dissociation from stereotypes may be rooted in other foundations.

Secondly, our findings indicate that the use of stereotypes for gender-based market segmentation may face societal resistance, particularly highlighted in the context of the #MeToo movement. However, the applicability of these results might be confined to this specific social context. Establishing a more general case on the effectiveness of gender-based segmentation and targeting strategies in the presence of stereotypical content warrants further exploration. Further research may use data on advertising agencies and advertising content to explore how marketers' own beliefs contribute to the creation of more or less stereotyped content. This investigation could explore the extent to which these strategies lose their effectiveness, particularly in times or locations where customers respond negatively to stereotypes.

Acknowledgments
The authors thank Anna Denisova, Jeanne Roboh, and Ye Jin Park for their research assistance. The authors also thank Anne Boring, Eve Colson-Sihra, and Murat Tunc for their helpful suggestions and comments, along with seminar participants at the Advances with Field Experiments 2022 Conference (University of Chicago), Marketing Science Conference 2023, Marketing in Israel (MII) Conference 2021, INSEAD, HEC, Hebrew University in Jerusalem, Tel Aviv University, Erasmus University Rotterdam, and LaLonde Conference 2019.

Appendix
Figure A.1 gives a detailed chronology of the #MeToo movement. Figure A.2 compares the evolution of search rates for sexual harassment topics six weeks before and after October 15, 2017, in exposed countries relative to unexposed countries. Figure A.3 plots the estimated DDD coefficient for each placebo date, alongside the baseline estimate with the true starting date. Table A.1 provides alternative triple-difference specifications.

Figure A.1. (Color online) #MeToo Movement: Timeline of Events

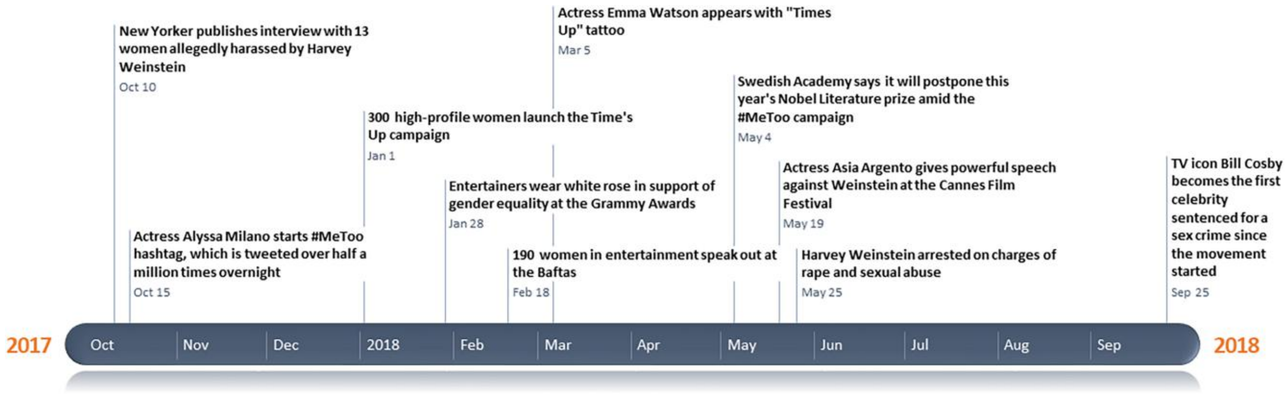
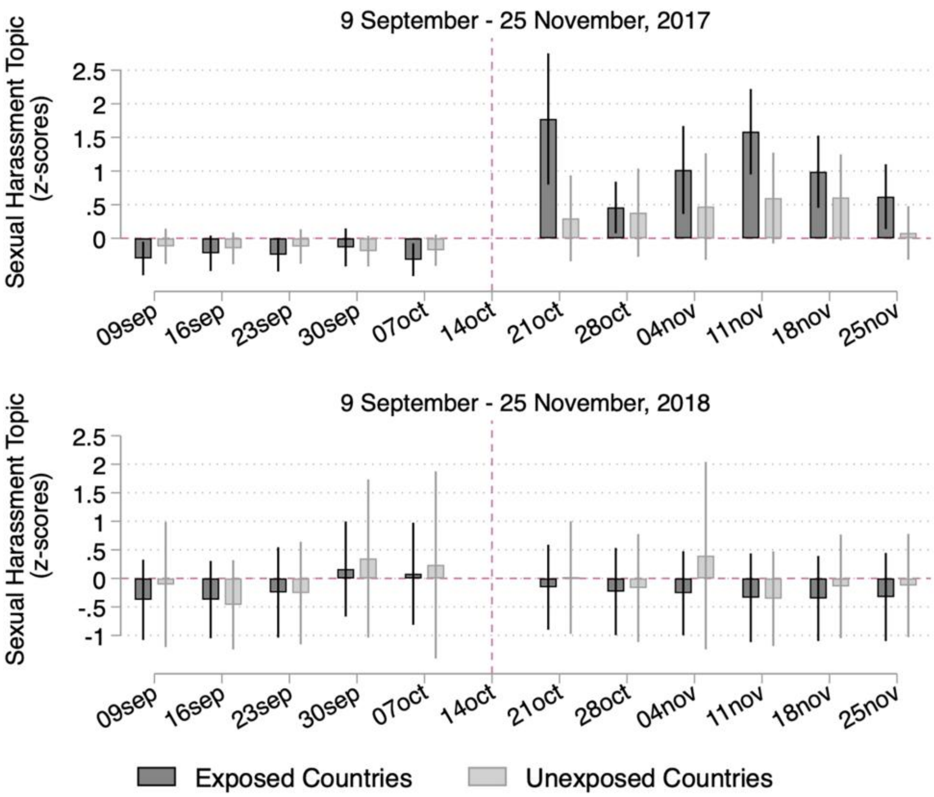
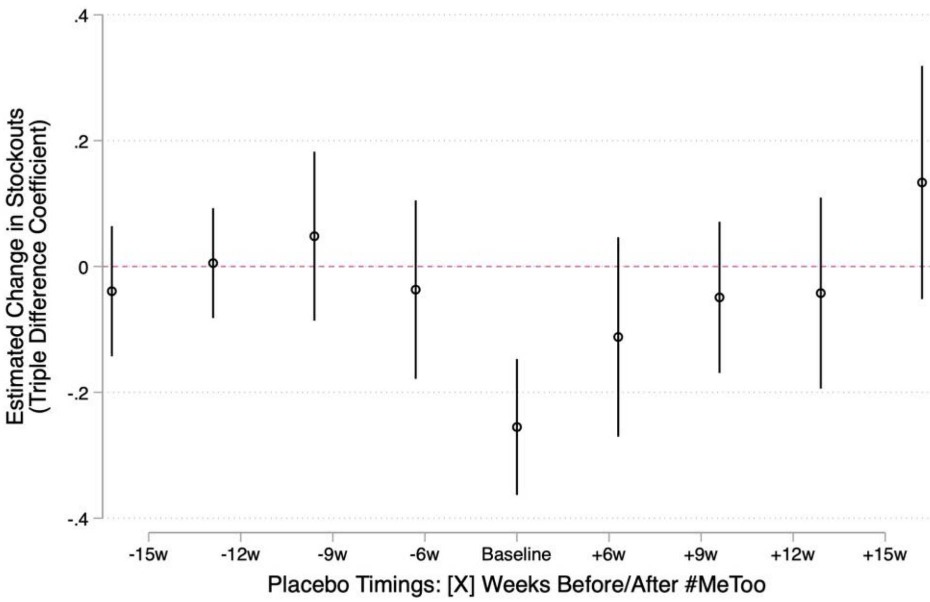


Figure A.2. Sexual Harassment Searches Around October 15 (2017 vs. 2018)



Notes. (Upper panel) Search volumes on sexual harassment topic (normalize using z-scores) six weeks before and six weeks after October 15, 2017, in unexposed countries (bottom tercile number of #MeToo tweets per million users and #MeToo headlines per million residents) vs. exposed countries (other countries). (Lower panel) Replication of the analysis using 2018 as a placebo year.

Figure A.3. Placebo Tests on the Start Date of the #MeToo Movement



Notes. Within-product-country DDD coefficients (together with their 95% confidence intervals), estimated using Poisson quasi-maximum likelihood on weekly product-level stockouts. All models control for the log of product prices and include product-country and country-week fixed effects. Each dot represents a separate regression estimated 6 weeks before (vs. 6 weeks after) the movement, where the movement is assumed to have started 6, 9, 12, or 15 weeks before or after its true "baseline" date (i.e., October 15, 2017). Robust standard errors are clustered at the country level.

Table A.1. Alternative Triple-Difference Specifications

	(1)	(2)	(3)
Panel A. Women's sample			
Post × #MeToo		0.098**	
		(0.042)	
Post × #MeToo × Nonstereotypical	0.098**		
	(0.042)		
Post × #MeToo × Stereotypical	−0.161***	−0.259***	−0.255***
	(0.051)	(0.058)	(0.055)
Observations	107,912	107,912	107,912
Panel B. Women's + men's sample			
Post × #MeToo		0.062*	
		(0.034)	
Post × #MeToo × men's shoes	0.062*		
	(0.034)		
Post × #MeToo × Nonstereotypical	0.097**	0.035	0.029
	(0.043)	(0.065)	(0.064)
Post × #MeToo × Stereotypical	−0.162***	−0.223***	−0.228***
	(0.051)	(0.062)	(0.057)
Observations	157,352	157,352	157,352
Product × Country FEs	Yes	Yes	Yes
Country × Week FEs	—	—	Yes

Notes. Within-product DDD estimators, estimated using Poisson quasi-maximum likelihood on weekly product-level stockouts. Panel A uses the sample of women's shoes only and uses nonstereotypical shoes as the reference category. Panel B includes both women's and men's shoes and uses men's shoes as the reference category. All models control for the log of product prices and include product-country fixed effects. Column 3 adds country-week fixed effects. Robust standard errors clustered at the country level are in parentheses.
$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Endnotes

[1] The lack of research-backed causal evidence contrasts with anecdotal evidence on the topic. For instance, #Movember—a movement exhorting men to grow facial hair to raise awareness of male-specific cancers—was reported to trigger a fall in sales of P&G's Gillette products even in the absence of specific calls to boycott them (Oldstone-Moore 2015, p. 1).

[2] The term was first used by Tarana Burke in 2006 to raise awareness about, and fight, sexual abuse and harassment against women. See Figure A.1 in the appendix for a detailed chronology.

[3] The recent rise of gender-neutral designs (or "unisex") in the fashion industry has also been discussed as evidence of a societal shift toward inclusiveness and the deconstruction of long-standing gender norms (Friedman and Trebay 2021).

[4] In essence, within the domain of women's footwear, products can exhibit feminine, neutral, or masculine characteristics. This contrasts with categories such as dresses or bras, where the gender associations are more universally feminine.

[5] Granger tests, where we exploit lags in daily tweets and online searches within countries, confirm the robustness of the effect.

[6] It is worth noting that #MeToo highlighted victim blaming as a root cause of harassment, which could have reduced, rather than intensified, the stigma associated with self-objectification.

[7] Previous research on social movements has mainly focused on their political aspects (e.g., emergence and organization (García-Jimeno et al. 2022)) and on specific consequences on business practices, regulations, or explicit calls for action, such as brand or product boycotts (Sun et al. 2021, Liaukonytė et al. 2023).

[8] Whereas Luo and Zhang (2021a) focus on the "production" of gender stereotypes (produced movie scripts), this paper investigates the demand side of the market, by focusing on the purchase of stereotypical products given a set of available products.

[9] In 75% of cases (out of a total of 2,886 product labeling tasks), research assistants indicated that the product or package color helped them to determine the gender target.

[10] In 2011 alone, women spent $38.5 billion on shoes in the United States alone, of which more than half went to heels (NPD Report 2011, http://www.npd.com/). In the United Kingdom alone in 2016, 37% of British female consumers bought at least one pair of heels (http://www.bbc.com/news/uk-37517468), making them the third most popular type of shoes after flat shoes (51%) and trainers (37%).

[11] White and Argo (2009) also show that women with low self-esteem confronted with an identity threat are more likely to choose a gender-neutral magazine (*Us Weekly* magazine) over a traditionally feminine one (*Cosmopolitan*).

[12] Kim et al. (2023) show that a calculator targeted at women through the label "For Women" is less likely to be chosen by women when the product is gender stereotypical (e.g., a purple calculator) relative to nonstereotypical (a green calculator).

[13] Of note, higher group status or self-esteem may also lead to a positive "pride" effect, which could increase identification to identity-related products (Ellemers et al. 1999).

[14] The data were obtained from StyleSage (now part of Centric Software), a market leader in global fashion analytics and trend forecasting.

[15] Data were not available for four OECD countries (Australia, Chile, Luxembourg, and New Zealand), because of the retailer's absence or marginal market penetration in these countries at that time.

[16] For robustness, we also obtained data on men's shoes (1,275 products), along with three other product categories within the women's section: lingerie products (168 products), dresses (4,135 products), and handbags (2,039 products).

[17] Full details on survey design and analysis can be found in Online Appendix B.

[18] On that date, actress Alyssa Milano started using the hashtag #MeToo, which was tweeted over half a million times overnight (see Figure A.1 for a detailed timeline of events). Consistent with other papers studying this movement (Lins et al. 2020, Levy and Mattsson 2023), we use October 15, 2017, as the pivotal date throughout.

[19] Following previous literature (Halberstam and Knight 2016, Gans et al. 2021), locations were extracted manually from the names marked by users as their current locations. For instance, the user may have written "Lima, Peru," which we marked as "Peru," because the name of the country is contained in the user's self-written location.

[20] We ascertain the count of Twitter users within each country using individuals accessible through Twitter advertisements in the year 2019 (Hootsuite 2019, Levy and Mattsson 2023).

[21] We observed a strong correlation ($r = 0.76$) between the number of residents and the average circulation of daily newspapers (in millions) in the 27 OECD countries for which such data were available (https://doi.org/10.1787/9789264088702-en). Although a more straightforward normalization approach could involve using the total number of headlines, Factiva does not provide information on the complete repository of articles for each country.

[22] See http://www.google.com/trends.

[23] For instance, keywords within the topic "sexual harassment" in 2017 include "*balance ton porc*" (the French equivalent of #MeToo) in France, whereas in Sweden, they include "Martin Timell" (a famous Swedish television presenter publicly accused in October 2017 of having sexually assaulted women).

[24] Sweden leads the ranking primarily because of a highly publicized case involving television presenter Martin Timell, who faced sexual abuse accusations as early as October 20, 2017, in the wake of the #MeToo movement. All results remain unchanged when removing Sweden from the sample.

[25] Levy and Mattsson (2023) also employ a triple-difference approach when estimating a causal impact of #MeToo on reporting sexual crimes over time, across countries, and between crime types.

[26] In other words, any product or country-level characteristics that do not change over time are absorbed by the product-country fixed effects, which is why Equation (1) does not include the *Stereotypical* and #*MeToo* variables (or their interaction).

[27] Like us, Levy and Mattsson (2023) estimate an ATT on a set of OECD countries. However, they rely on quarterly reporting of sex crimes whereas we use high-frequency weekly and daily stockout data. Their main measure of #MeToo exposure is based on Google search data. They also perform a battery of robustness tests, among which they use #MeToo Tweets, and a survey-based measure on a smaller set of 12 OECD countries as a measure of exposure.

[28] When defining stereotypical shoes as possessing at least one out of three markers, about half of the women's shoes get classified as "stereotypical." For robustness, we also report results using this alternative definition.

[29] We also show results where we choose a different threshold to define unexposed countries (bottom quartile; 8 countries), use a dummy of "weakly exposed" countries (below median; 16 countries), or use the index as a continuous measure of treatment intensity.

[30] Product prices themselves may be affected by #MeToo, for instance, if retailers adjusted the prices of certain stereotypical products downward to avoid a drop in stockouts. Such pricing strategies could, theoretically, introduce a downward bias in our main triple-difference estimate when controlling for price changes.

[31] We do not cluster within weeks given the small number of clusters. Using Wild bootstrap clustering at the country and week levels with *boottest*, we find a somewhat larger p-value on the three-way interaction coefficient ($p = 0.015$). All other coefficients remain nonsignificant with p-values higher than 0.394.

[32] Given the absence of exogenous variation in price, the η parameter may not capture the true price elasticity of demand. For instance, retailers may decide to cut prices when they anticipate lower demand, which would bias the parameter downward.

[33] When looking at supply-side effects, our results confirm that the #MeToo movement had no immediate effect on product pricing (Table 4).

[34] As discussed in Section 5.5, our estimate is closer to Levy and Mattsson (2023) if we use their definition of "strong" exposure (i.e., above median exposure).

[35] As the composite index also incorporates search data, we use the number of tweets per million users and the number of #MeToo headlines per million residents to define exposure here.

[36] We use z-scores so that the *Post* and *Post* × *Stereotypical* coefficients report the pre-post change in stockouts at the mean of #MeToo exposure and their interaction with #*MeToo* reports the effect of a one-standard-deviation increase in exposure ($\mu = 0.25$, $\sigma = 0.19$).

[37] We observe a smaller point estimate when we use #MeToo tweets per million users as the measure of exposure.

[38] We use the estimated parameters obtained through regression (4) in Online Appendix B.

[39] The "stereotypical" products identified using the presence of at least two markers boast an average stereotypical index of 6.2 (indicating a classification between "feminine" and "very feminine"), compared with an average of 4.9 for the other products (falling between "neutral" and "somewhat feminine").

[40] The gender-stereotypical association of colors is captured by the α_j coefficients in regression (4) in Online Appendix B. Neutral color is the excluded reference color.

[41] Levy and Mattsson (2023) present evidence suggesting that the impact of #MeToo on the reporting of sexual offenses persisted for at least two years. The current context makes it difficult to examine long-term outcomes, because of potential responses from the supply side.

[42] Men's shoes exhibit considerably less variation on this index, driven primarily by color gender associations because of limited variation in styles and the absence of heels.

[43] We use 2018 as out-of-stock sizes were not systematically recorded by the data provider in 2016.

[44] It is important to interpret these results cautiously because of potential retailer adaptation to the #MeToo shock in stockout predictions a year later. However, the absence of a "rebound" effect around the first anniversary of the #MeToo movement adds credibility to the validity of the approach (Figure A.2).

[45] As documented earlier, nearly 95% of observations cover products available in at least 26 OECD countries in the period considered.

[46] The scale is based on the following question: "This style of shoe would catch the attention of men if it were worn by a woman. Fully disagree (1), Disagree (2), Somewhat disagree (3), Neutral (4), Somewhat agree (5), Agree (6), Fully agree (7)." See Online Appendix B. Although both measures are correlated, their associated attributes differ. For example, "pink" is about eight times more gender stereotypical than "red," whereas "red" is nearly three times more likely to attract men's attention compared with "pink." Regarding product shapes and their appeal to men, boots rank the highest, with

less variation among other shapes. Heel height is the only attribute yielding similar results.

[47] Consumers may also walk away from items that used to serve as a substitute for bolstering one's sense of power, such as high heels. Indeed, high heels are also known for adding height and can be perceived as a way to boost one's stature and, in turn, sense of power.

[48] This treatment also fails at improving stated sense of power, however.

[49] We use the following GPT prompt: "Which of these five emotions—happiness, anger, surprise, sadness, fear—best represents the mental state of a person writing the following English text? Answer only with a number: 1 if happiness, 2 if anger, 3 if surprise, 4 if sadness, 5 if fear." There is no major difference in the occurrence of surprise across conditions.

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